

ENGLISH FOR BUILDING COMFORT AND REFRIGERATION TEAMS

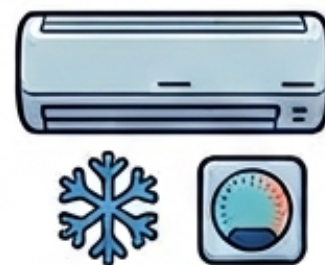
HVAC & Refrigeration English

Clear readings. Reliable handovers.

Phil Stilwell

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For heating, ventilation, air-conditioning, and refrigeration staff communicating with clients, facilities teams, and project partners.

2026 edition

Eight scenarios. Eight skills.

Eight conversations develop precise reporting and coordination: clarify a thermostat value, reconcile equipment labels, explain supply and return air, compare ratings, report moisture, escalate a cabinet alarm, discuss a grille move, and hand over unfinished commissioning items.

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Twenty-two is the target, not the finding | Separate a thermostat setpoint from a measured room temperature while preserving the time and location of a comfort complaint.

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06 Discussing refrigeration alarms and temperature records

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An alarm, a display value, and missing product data | Report an alarm chronology accurately, distinguish cabinet and product information, and route a food-safety question to the responsible lead.

07 Coordinating ductwork, access, and changes

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Moving a grille is a coordinated change | Acknowledge an appearance preference, explain unassessed consequences, and present two bounded choices without promising price or timing.

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Keep the reading attached to its meaning.

Technical English in this field depends on what a number measures, which component a label identifies, and what a record actually establishes. Practice explaining those distinctions to colleagues and customers.

Name the value before interpreting it

A thermostat setpoint is a target, not automatically the measured room temperature. Cooling capacity and electrical input can both use kilowatts but describe different quantities. Keep the quantity name and unit together.

"The 22 degrees Celsius is the setpoint; no ambient reading has been supplied."

Preserve component identity

Similar suffixes on an outdoor unit and an air handler do not prove a verified pairing. Use the project register and confirmed references rather than treating a familiar shorthand label as the whole system.

"CU-4 and AH-4 appear in separate records; their relationship needs confirmation."

Separate history, findings, and decisions

A past moisture report can coexist with a dry floor during a later visit. A cabinet display and a defrost entry do not establish food temperatures or release food for service. Name the evidence and the responsible decision route.

"The display recorded 8 degrees Celsius; product temperatures and event duration are not supplied."

Keep unfinished handover items visible

Receiving a commissioning report does not complete a deferred check. Correct documents, scheduling responsibility, and update dates need separate entries. A named document owner does not automatically own every outstanding action.

"Nia owns document correction; responsibility for scheduling the seasonal check remains open."

PROFESSIONAL BOUNDARIES

The projects, products, values, and records are fictional. This book teaches English, not equipment operation, diagnosis, installation, refrigerant handling, food-safety decisions, or compliance. Follow actual training, authorization, manufacturer instructions, site procedures, and applicable requirements. No dialogue authorizes roof access, testing, adjustment, refrigerant work, food release, or a system alteration. Equipment readings and document receipt do not establish safety or overall performance.

Twenty-two is the target, not the finding

A Read the situation

Separate a thermostat setpoint from a measured room temperature while preserving the time and location of a comfort complaint.

Office manager Asha tells service coordinator Ben that the meeting room is 22 degrees Celsius. The number is actually the thermostat setpoint, not a confirmed ambient-temperature reading. Staff report feeling warm there after 14:00. No cause or equipment fault has been established. Ben needs to clarify the value, keep the staff report attributed, and record the afternoon pattern. The conversation uses existing information and gives no instruction to adjust or test the equipment.

PEOPLE / Asha: Office manager | Ben: Service coordinator

CORRECT THE NUMBER LABEL WITHOUT DISMISSING THE PEOPLE

A setpoint can be accurate while a comfort complaint is still meaningful. Ask what the displayed value represents before treating it as a measurement. Acknowledge the occupants and preserve their reported pattern without diagnosing a fault or prescribing a control change.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does 22 degrees Celsius represent?

- | | |
|---|-----------------------------------|
| A An outdoor temperature | B The thermostat setpoint |
| C A verified ambient-temperature reading | D A supply-air measurement |

2. When do staff report feeling warm?

- | | |
|--|---|
| A Exactly at a confirmed failure time | B Throughout every day without exception |
| C After 14:00 | D Only before 08:00 |

3. What is established about the equipment?

- | | |
|---|---|
| A The system is confirmed fault-free | B No cause or fault has been established |
| C The thermostat sensor has failed | D The cooling capacity is inadequate |

Answers and explanations: [page 85](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

HVAC

Heating, ventilation, and air conditioning.

IN USE discuss the HVAC system

thermostat

Control associated with maintaining a selected temperature.

IN USE identify the thermostat

setpoint

Target value used by a control system.

IN USE distinguish the setpoint

ambient temperature

Temperature of the surrounding air at the relevant location.

IN USE record ambient temperature

temperature reading

Value reported by a temperature-measuring device.

IN USE identify the temperature reading

degrees Celsius

Temperature unit used in the supplied report.

IN USE state the value in degrees Celsius

comfort complaint

Occupant report that environmental conditions feel uncomfortable.

IN USE log a comfort complaint

occupant feedback

Account supplied by people using the space.

IN USE preserve occupant feedback

sensible cooling

Heat removal associated with lowering air temperature without counting moisture condensation.

IN USE distinguish sensible cooling

afternoon pattern

Reported tendency occurring later in the day.

IN USE record the afternoon pattern

sensor

Device responding to a physical condition for measurement or control.

IN USE identify the relevant sensor

measured value

Quantity established by a measurement.

IN USE distinguish the measured value

Precision check

Twenty-two degrees Celsius is the setpoint in this case. It must not be relabeled as ambient temperature. The complaint concerns staff feeling warm in the meeting room after 14:00, not a verified temperature or fault.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

displayed value

Number shown by a device or interface.

IN USE clarify the displayed value

target temperature

Temperature the control is intended to seek.

IN USE state the target temperature

latent cooling

Heat removal associated here with condensing water vapor from air.

IN USE account for latent cooling

relative humidity

Water-vapor level relative to saturation at the same temperature.

IN USE report relative humidity

occupancy

Use of a space by people.

IN USE describe room occupancy

control schedule

Timing arrangement for control settings or operation.

IN USE review the control schedule

zone

Defined area treated as a control or service region.

IN USE identify the relevant zone

heat gain

Heat added to a space from its surroundings or internal sources.

IN USE assess heat gain

equipment fault

Established malfunction of a system component.

IN USE avoid assuming an equipment fault

reported period

Time span described in an account.

IN USE preserve the reported period

service description

Concise account used to request relevant attention.

IN USE clarify the service description

attributed report

Statement linked to the person or group who supplied it.

IN USE retain an attributed report

Precision check

A comfort report is useful even without a confirmed ambient reading. Retain the people, place, and afternoon pattern. Do not diagnose a sensor failure, prescribe a setting change, or claim normal performance from the target alone.

Phrases you can actually use

C Notice the language / 1 of 2

ASK WHAT THE NUMBER MEANS

"Is 22 the setpoint or a measured room temperature?"

PRESERVE THE MISSING READING

"No confirmed ambient reading has been supplied."

KEEP THE TIME PATTERN

"The reports concern the period after 14:00."

SEPARATE THE EVIDENCE

"We have a setpoint and an occupant report, not a diagnosis."

NAME THE TARGET

"The thermostat is set to 22 degrees Celsius."

ACKNOWLEDGE THE PEOPLE

"Staff report feeling warm in the meeting room."

AVOID A CONTRADICTION

"A target setting does not disprove a comfort complaint."

AVOID ASSIGNING A FAILED PART

"No sensor fault has been established."

Set to

Describes a target, not necessarily a measured condition.

Report feeling

Attributes a subjective experience without dismissing it.

After 14:00

Preserves the reported period without inventing an exact onset.

Phrases you can actually use

C Notice the language / 2 of 2

ASK ABOUT EXISTING INFORMATION

"What does the current record identify that value as?"

USE THE CORRECT UNIT

"The value is stated in degrees Celsius."

RETAIN THE ACCOUNT

"I will record what the staff have reported."

READ BACK THE DISTINCTION

"Setpoint 22; staff feel warm after 14:00; ambient reading unavailable."

KEEP THE LOCATION

"Please keep meeting room in the service description."

AVOID PRESCRIBING AN ADJUSTMENT

"This clarification does not specify a new setting."

AVOID INVENTING A MEASURED EXCESS

"We do not have a measured temperature above the target."

CLOSE WITH AN ACCURATE REQUEST

"I will pass on that description for the relevant assessment."

Not supplied

Means information is absent from the account, not that measurement is impossible.

Does not disprove

Allows the setpoint and occupant experience to coexist.

No fault established

Avoids both a diagnosis and an unsupported fault-free assurance.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which wording accurately uses the number?

- A** The thermostat setpoint is 22 degrees Celsius
- B** The room was measured at exactly 22
- C** Supply air is confirmed at 22
- D** The outdoor temperature caused the problem at 22

2. Which summary preserves the complaint?

- A** A sensor fault begins at 14:00
- B** Staff feel warm in the meeting room after 14:00; ambient reading not supplied
- C** Staff are mistaken because the setting is 22
- D** Every room overheats all day

3. What should Ben avoid inferring?

- A** The reported room
- B** The afternoon pattern
- C** A particular equipment fault
- D** The value's unit

4. Which question best clarifies the displayed value?

- A** Does the record label 22 as the target or the measured room temperature?
- B** Why is the compressor broken?
- C** Which new setting should we choose immediately?
- D** Can the client test the equipment now?

[Answers and explanations: page 85.](#)

Twenty-two is the target, not the finding

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

equipment fault / attributed report / target temperature / occupant feedback / temperature reading / setpoint / ambient temperature / afternoon pattern / degrees Celsius / service description

- 01 **Asha:** Ben, the meeting room is twenty-two degrees, but staff keep saying it feels warm after lunch. How should I describe that without sounding contradictory?
- 02 **Ben:** Before I record it, is twenty-two the ¹ _____, or is it a separate measured room-temperature value? Those mean different things in a service request.
- 03 **Asha:** It's the thermostat setting, not a separate room measurement. I copied the number from the information the office team gave me.
- 04 **Ben:** Then we should not call it the ² _____. We can record the target and the staff account without treating the setting as proof of the room condition.
- 05 **Asha:** The reports concern the meeting room after two in the afternoon. I don't have the same complaint about every room or every part of the day.
- 06 **Ben:** That is a useful ³ _____. I will retain the meeting-room location and after 14:00, so the request does not become a whole-office complaint.
- 07 **Asha:** Will the service team dismiss the complaint if the request says the thermostat is set to twenty-two? The people still feel uncomfortable.
- 08 **Ben:** No. We can preserve the ⁴ _____ and clarify the number at the same time. A target does not establish what everyone in the room actually experiences.
- 09 **Asha:** I had written possible sensor failure to make the request more specific. But the information doesn't establish a failed sensor, does it?
- 10 **Ben:** It is. No ⁵ _____ has been established. We should not name a failed sensor or another component simply to give the complaint a technical-sounding cause.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Asha:** Please include Celsius. Our overseas colleagues sometimes use Fahrenheit, and I'd rather the unit didn't disappear when the request is forwarded.
- 12 **Ben:** Certainly. I will state ⁶ _____ explicitly. A number without its unit and its meaning can become misleading when the message is forwarded.
- 13 **Asha:** When we say no confirmed reading supplied, we mean this account lacks it. We don't know whether someone has measured the room on another occasion.
- 14 **Ben:** Yes. The missing ⁷ _____ is a limit of this information. It does not establish the history of every measurement or the condition of the equipment.
- 15 **Asha:** I'll keep the request to the existing information. I don't want my summary mistaken for an instruction to change the thermostat or test a component.
- 16 **Ben:** Agreed. We are clarifying a ⁸ _____, not issuing control instructions. Any assessment belongs with the relevant team and its actual procedures.
- 17 **Asha:** Could you read it back with the target and missing measurement separate? I need a short version for the facilities manager.
- 18 **Ben:** Here is the ⁹ _____: staff feel warm in the meeting room after 14:00; thermostat setpoint 22 degrees Celsius; no confirmed ambient reading supplied; cause unestablished.
- 19 **Asha:** That preserves the staff account without inventing a measured temperature above target. I'll stop describing twenty-two as the actual room condition.
- 20 **Ben:** I will pass on that record. It distinguishes the ¹⁰ _____ from the actual condition that still needs assessment, while keeping the staff's reported experience visible.

ANSWERS / EXPLANATIONS: PAGE 86

1. setpoint | 2. ambient temperature | 3. afternoon pattern | 4. occupant feedback | 5. equipment fault | 6. degrees Celsius | 7. temperature reading | 8. service description | 9. attributed report | 10. target temperature

Label the value correctly

F Say it

Complete the service report using the target, time pattern, and missing measurement. Do not add a diagnosis.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

14:00 / setpoint / ambient / unestablished

Manager: "The thermostat ¹ _____ is 22 degrees Celsius."

Coordinator: "Staff feel warm after ² _____."

Manager: "No confirmed ³ _____ reading is supplied."

Coordinator: "The cause remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 86

1. setpoint | 2. 14:00 | 3. ambient | 4. unestablished

Rehearse with a fixed script

1. Read turns 1-10. Stress setpoint, no ambient reading, meeting room, and after two to preserve the complaint's meaning.

2. Switch roles for turns 11-20. Keep Celsius with the target and distinguish missing information from no measurement ever taken.

3. Complete and check the service exchange. Read the actual report without inventing a sensor fault or setting change.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

AC-4, CU-4, and AH-4 are not yet a pairing

A Read the situation

Reconcile conflicting equipment references using existing records and preserve an unverified relationship between components.

A school service request calls the system AC-4. The roof plan labels an outdoor unit CU-4, while the indoor equipment register identifies air handler AH-4. Their relationship is not confirmed. Facilities manager Mira has the equipment register and discusses the request with service coordinator Owen. They need to preserve the three source references and seek confirmation through the records process. Neither person needs to access the roof or inspect equipment to clarify the existing documents.

PEOPLE / Mira: Facilities manager | Owen: Service coordinator

TREAT A FAMILIAR LABEL AS A STARTING POINT

Site shorthand may name a whole system while formal documents identify separate components. Similar numbering can suggest a question, but it is not proof of connection. Cite each label with its source so the next person can reconcile the records without silently merging them.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which label appears in the service request?

- A AH-4 only
- C AC-4

- B A verified combined asset number
- D CU-4 only

2. What does the register identify as AH-4?

- A An indoor air handler
- C A roof-access permit

- B A confirmed outdoor condenser
- D A room thermostat setting

3. What is unresolved?

- A Whether Mira has the register
- C Whether the roof plan contains CU-4

- B Whether the request uses AC-4
- D The relationship between CU-4 and AH-4

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

equipment register

Record identifying equipment and relevant asset information.

IN USE consult the equipment register

asset tag

Assigned identifier used to distinguish an equipment item.

IN USE preserve the asset tag

system reference

Label used to identify an overall system.

IN USE clarify the system reference

component reference

Label identifying a particular part of a system.

IN USE retain the component reference

air handler

Equipment that moves and may condition air in a building system.

IN USE identify the air handler

outdoor unit

Equipment assembly located outside the served interior.

IN USE identify the outdoor unit

condensing unit

Assembly associated with refrigerant compression and heat rejection in relevant systems.

IN USE verify the condensing-unit reference

indoor unit

Equipment assembly located within the building or served space.

IN USE identify the indoor unit

equipment pairing

Confirmed relationship between components intended to operate together.

IN USE verify the equipment pairing

served area

Space associated with the service from an identified system.

IN USE confirm the served area

roof plan

Drawing showing relevant roof-level features and equipment.

IN USE cite the roof plan

record source

Document or system from which information was obtained.

IN USE identify the record source

Precision check

AC-4 appears in the request, CU-4 on the roof plan, and AH-4 in the indoor register. Keep all three with their sources. A shared numeral does not confirm a component pairing or served area.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

cross-reference

Link connecting corresponding entries in different records.

IN USE establish a cross-reference

label discrepancy

Difference between identifiers in related information.

IN USE record the label discrepancy

model number

Manufacturer identifier for a product design or version.

IN USE confirm the model number

serial number

Identifier distinguishing an individual manufactured unit.

IN USE record the serial number

nameplate

Manufacturer label carrying equipment identification or ratings.

IN USE use recorded nameplate information

asset hierarchy

Organized relationship between systems and their components.

IN USE clarify the asset hierarchy

service request

Record asking for attention to an identified issue or system.

IN USE reconcile the service request

records reconciliation

Process of resolving differences between documented information.

IN USE complete records reconciliation

unverified link

Relationship that has not been confirmed.

IN USE retain an unverified link

identification query

Request to clarify which equipment is meant.

IN USE raise an identification query

record custodian

Person responsible for maintaining or providing a record.

IN USE identify the record custodian

confirmed mapping

Verified correspondence between identifiers or components.

IN USE obtain confirmed mapping

Precision check

Mira already has the equipment register. The next step is records reconciliation through the responsible process, not asking someone to reach a roof or inspect a nameplate. Recorded information must still be verified in context.

Phrases you can actually use

C Notice the language / 1 of 2

QUOTE THE REQUEST

"The service request calls the system AC-4."

QUOTE THE REGISTER

"The indoor register identifies air handler AH-4."

AVOID A NUMBER SHORTCUT

"The shared four does not prove that the components are paired."

USE EXISTING RECORDS

"You already have the equipment register."

QUOTE THE PLAN

"The roof plan labels an outdoor unit CU-4."

PRESERVE SOURCE ATTRIBUTION

"Each label should stay attached to its source."

NAME THE UNRESOLVED LINK

"The relationship between CU-4 and AH-4 is unverified."

REQUEST RECONCILIATION

"We need a confirmed cross-reference between these entries."

Labels versus connects

A label names an item; it does not establish the item's relationship to another.

Appears in

Attributes an identifier to a particular source.

May not

Warns against assuming equivalence without declaring the records wrong.

Phrases you can actually use

C Notice the language / 2 of 2

SEPARATE SYSTEM AND COMPONENT

"A system label may not be the identifier of one physical unit."

KEEP THE SERVED AREA OPEN

"We have not confirmed the served area from these labels."

KEEP THE REQUEST PRECISE

"Please retain all three references in the query."

READ BACK THE GAP

"The records name the items but do not confirm their relationship."

AVOID ASSUMED RENAMING

"I will not replace AC-4 with CU-4 by guesswork."

DISTINGUISH IDENTIFIERS

"A model number and a serial number identify different things."

AVOID DIRECTING ACCESS

"This is a document query, not a request for roof access."

CLOSE WITH THE NEEDED RESULT

"We need confirmed mapping before merging the references."

By guesswork

Identifies an unsupported substitution of one label for another.

Already have

Directs attention to available information rather than new physical access.

Before merging

Makes verification a condition of consolidating records.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which summary is accurate?

- A** All three labels identify one physical item
- B** Request AC-4; roof plan CU-4; register AH-4; relationship unverified
- C** AC-4 is definitely another name for CU-4
- D** AH-4 and CU-4 are paired because both end in four

2. What should Owen ask Mira to provide?

- A** A revised request replacing AC-4 with CU-4 immediately
- B** A cross-reference based only on the shared numeral
- C** A served-area label borrowed from another request
- D** The relevant existing register information for reconciliation

3. Why is the numeral four insufficient?

- A** AH always means the entire outdoor system
- B** Similar numbering does not establish verified component relationships
- C** Numbers can never be used in asset labels
- D** All outdoor units must have different numerals

4. What should remain visible until clarified?

- A** The source of each label and the unverified link
- B** Only the shortest label
- C** A presumed served area
- D** An invented nameplate reading

Answers and explanations: [page 87](#).

AC-4, CU-4, and AH-4 are not yet a pairing

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

confirmed mapping / air handler / equipment pairing / served area / system reference / model number / service request / equipment register / record source / cross-reference

- 01 **Mira:** Owen, the school's request says AC-4, but I can't match that cleanly to the project documents. I don't want to send the wrong reference.
- 02 **Owen:** Let's keep the ¹ _____ beside each identifier. What does the roof plan call the outdoor equipment, and what does the indoor register say?
- 03 **Mira:** The roof plan labels the outdoor unit CU-4. The indoor register lists AH-4, an air handler. Only the service request uses AC-4.
- 04 **Owen:** So the ² _____ has its own reference. AC-4 may refer to the system more broadly; we shouldn't treat all three labels as one physical item.
- 05 **Mira:** They all end in four. I thought that would be enough to link the indoor and outdoor entries, but the records don't actually confirm it.
- 06 **Owen:** Then leave the ³ _____ unverified. The shared numeral suggests a question to resolve; it doesn't establish that those components operate together.
- 07 **Mira:** I have the register here already. Can we resolve the question through the records process instead of asking someone to go onto the roof?
- 08 **Owen:** Yes. Send the relevant ⁴ _____ entries and roof-plan reference. We're reconciling existing documents, not requesting physical access or a new equipment inspection.
- 09 **Mira:** Would it help if I changed AC-4 to CU-4 on the request now? At least that label would agree with the roof plan.
- 10 **Owen:** Please retain the original ⁵ _____ wording. Add the plan and register references alongside it; changing the label prematurely could narrow what the requester meant.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Mira:** Right, a request about the whole system might become one about just the outdoor unit. We need the relationship before choosing the final reference.
- 12 **Owen:** Exactly. A ⁶ _____ and a component identifier can describe different levels of the equipment arrangement. The records need a supported link between them.
- 13 **Mira:** There may be model and serial fields in the register too. Those aren't interchangeable with the school's asset tags, are they?
- 14 **Owen:** No. A ⁷ _____ identifies a product design or version; a serial number distinguishes a manufactured unit. Keep each field with its recorded equipment item.
- 15 **Mira:** Another request mentions the meeting room. Can I borrow that location to make this one more useful, or would that introduce an unsupported connection?
- 16 **Owen:** Don't borrow it. The ⁸ _____ still needs confirmation for these references. A plausible location from a different request isn't evidence for this equipment relationship.
- 17 **Mira:** Please phrase the query without saying one document must be wrong. The labels may describe different things correctly, even if their connection is missing.
- 18 **Owen:** Ask for a confirmed ⁹ _____ between request AC-4, plan item CU-4, and register item AH-4, including the component relationship and the served area.
- 19 **Mira:** I'll send all three sources intact. That should let the record owner investigate the actual mapping instead of guessing which label we preferred.
- 20 **Owen:** Once we receive ¹⁰ _____, we can clarify the service reference. Until then, keep the separate labels and unresolved relationship visible in the request.

ANSWERS / EXPLANATIONS: PAGE 88

1. record source | 2. air handler | 3. equipment pairing | 4. equipment register | 5. service request | 6. system reference | 7. model number | 8. served area | 9. cross-reference | 10. confirmed mapping

Keep all three references

F Say it

Complete the identification exchange using the three labels and the unresolved relationship. Use existing records only.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

AC-4 / unverified / CU-4 / AH-4

Manager: "The request says ¹ _____."

Coordinator: "The roof plan identifies the outdoor unit as ² _____."

Manager: "The register identifies the air handler as ³ _____."

Coordinator: "Their relationship remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 88

1. AC-4 | 2. CU-4 | 3. AH-4 | 4. unverified

Rehearse with a fixed script

1. Read turns 1-10. Say each identifier with its source: request AC-4, roof plan CU-4, indoor register AH-4.

2. Switch roles for turns 11-20. Contrast system, component, model, serial, and served area without merging those fields.

3. Complete and check the identification exchange. Retain the unverified relationship rather than pairing items by their numeral.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Two ceiling openings, two functions

A Read the situation

03

Translate supply and return labels into plain language while preserving the difference between a comfort report and an airflow measurement.

Client Tomas calls both ceiling openings air-conditioning outlets. The current drawing identifies S3 as supply air and R2 as return air. He reports discomfort below S3 but supplies no airflow measurements. Technician Leila explains the labels and records the location of the complaint. The drawing establishes the stated functions, not measured performance. No cause, balancing result, or adjustment is supplied, and the conversation must not prescribe changes to grilles or controls.

PEOPLE / Tomas: Client | Leila: Technician

TRANSLATE THE FUNCTION, THEN RETURN TO THE COMPLAINT

An everyday label can conceal an important functional difference. Explain the direction of airflow simply, then repeat the actual complaint. Technical vocabulary should improve the report, not overwhelm the customer or imply a diagnosis that the measurements do not support.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which opening does the drawing identify as supply?

A Neither opening

B S3

C R2

D Both openings

2. Where does Tomas report discomfort?

A At an identified outdoor unit

B Below S3

C Below R2 only

D In every room

3. What is missing?

A The existence of two openings

B The client's location description

C Airflow measurements

D The drawing labels

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

03

Read the definitions and say the common word combinations aloud.

supply air

Air delivered into a served space through the distribution system.

IN USE identify supply air

return air

Air drawn from a space back toward the air-handling system.

IN USE identify return air

supply grille

Grille associated with delivery of air into the space.

IN USE locate the supply grille

return grille

Grille associated with air returning from the space.

IN USE identify the return grille

diffuser

Air-distribution device designed to spread delivered air.

IN USE identify the diffuser

register

Grille assembly that may include an airflow-control damper.

IN USE distinguish the register

duct

Passage used to convey air.

IN USE identify the duct

ductwork

Network of ducts and associated components.

IN USE review the ductwork

air distribution

Delivery and movement of air within a served area.

IN USE describe air distribution

airflow rate

Volume of air passing a point per unit time.

IN USE report the airflow rate

cubic feet per minute

Volume-flow unit commonly abbreviated CFM.

IN USE state cubic feet per minute

liters per second

Volume-flow unit used for air or liquid movement.

IN USE record liters per second

Precision check

S3 is labeled supply and R2 return on the current drawing. This tells the reader their stated functions. It does not establish actual flow rate, velocity, balance, or the cause of the discomfort.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

air velocity

Speed of air movement at a stated position.

IN USE distinguish air velocity

air changes per hour

Hourly airflow divided by room volume, with the air source specified.

IN USE distinguish total and outdoor-air changes

draft

Air movement perceived as unwanted by an occupant.

IN USE describe a reported draft

occupied zone

Part of a room normally used by people.

IN USE identify the occupied zone

throw

Distance associated with a defined air jet under stated conditions.

IN USE review diffuser throw

spread

Lateral extent of a delivered air stream under defined conditions.

IN USE distinguish spread from throw

damper

Component used to regulate or control airflow in a system.

IN USE identify the damper

balancing

Technical adjustment and verification of distribution to meet specified requirements.

IN USE distinguish balancing from explanation

design airflow

Airflow value specified by the design.

IN USE compare design airflow

measured airflow

Airflow established through the relevant measurement.

IN USE record measured airflow

distribution report

Account describing air delivery or related comfort observations.

IN USE clarify the distribution report

performance finding

Supported conclusion about how a system functions.

IN USE distinguish a performance finding

Precision check

The report concerns discomfort below S3. Preserve that location without changing it to a measured excess or deficit of airflow. Explanation of the labels is not an instruction to adjust a grille, damper, or control.

Phrases you can actually use

C Notice the language / 1 of 2

03

NAME THE TWO FUNCTIONS

"S3 is supply air; R2 is return air."

EXPLAIN RETURN

"Return air goes back toward the air-handling system."

KEEP THE COMPLAINT LOCATION

"You report discomfort below S3."

SEPARATE RATE AND SPEED

"Airflow rate and air velocity are different quantities."

EXPLAIN SUPPLY

"Supply air is delivered into the room."

ATTRIBUTE THE LABELS

"Those are the functions shown on the current drawing."

AVOID CONVERTING EXPERIENCE INTO MEASUREMENT

"We do not have an airflow measurement for that location."

NAME COMMON UNITS

"Volume flow may be stated in cubic feet per minute or liters per second."

Shown on the drawing

Attributes the stated function without verifying installed performance.

Into versus back toward

Gives the customer a simple directional contrast.

Below S3

Preserves the complaint location rather than generalizing to the whole system.

Phrases you can actually use

C Notice the language / 2 of 2

03

AVOID A BALANCE CLAIM

"The labels do not establish that the system is balanced."

CLARIFY THE EVERYDAY TERM

"Calling both openings outlets hides their different functions."

AVOID PRESCRIBING A CHANGE

"This explanation does not specify an adjustment."

DISTINGUISH DESIGN FROM RESULT

"A design value is not automatically a measured result."

KEEP CAUSE OPEN

"The cause of the discomfort has not been established."

RETAIN THE CUSTOMER'S WORDS

"I will record the discomfort without inventing a measured excess."

ASK FOR THE EXISTING REFERENCE

"Please keep S3 attached to the reported location."

CLOSE WITH A PRECISE REPORT

"Discomfort below supply S3; airflow data not supplied; cause unresolved."

Not automatically

Blocks an unsupported equivalence between design and measurement.

Reported discomfort

Records the experience without assigning a technical cause.

Does not specify

Keeps an explanation separate from an operating instruction.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which explanation is accurate?

- A** Supply means measured excess airflow
- B** Supply delivers air into the room; return carries air back toward the system
- C** Both openings must deliver air into the room
- D** Return means outdoor exhaust in every system

2. What can the drawing labels establish here?

- A** The cause of the discomfort
- B** The correct damper adjustment
- C** The stated functions of S3 and R2
- D** A verified airflow rate

3. Which report invents a measurement?

- A** Tomas reports discomfort below S3
- B** R2 is labeled return
- C** No airflow data were supplied
- D** S3 delivers too many liters per second

4. What is the proper scope of Leila's response?

- A** Prescribe a grille adjustment
- B** Declare the system balanced
- C** Reject the complaint because S3 is supply
- D** Explain labels and preserve the complaint for assessment

Answers and explanations: [page 89](#).

Two ceiling openings, two functions

E Conversations / turns 1-10

03

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

measured airflow / performance finding / damper / occupied zone / balancing / distribution report
/ Air velocity / supply air / return air / cubic feet per minute

- 01 **Tomas:** Leila, the ceiling outlet above this seat is uncomfortable to sit under. It's S3 on the drawing. The other outlet is R2.
- 02 **Leila:** S3 is labeled ¹ _____ on the drawing. R2 has a different function, so calling both outlets can hide a useful distinction.
- 03 **Tomas:** They look like parts of the same system. Does R2 also send air into the room, just from a different position?
- 04 **Leila:** R2 is labeled ² _____. Supply delivers air into the space; return carries air back toward the air-handling system. Return isn't automatically an outdoor exhaust.
- 05 **Tomas:** Then the complaint belongs below the supply opening, S3. Please keep that location rather than writing that the whole room's ventilation has failed.
- 06 **Leila:** I'll retain it in the ³ _____: discomfort below S3. We can identify the drawing label without claiming that it verifies how the installed system performs.
- 07 **Tomas:** I was about to say S3 delivers too much air. That's how it feels here, but I don't have a measured figure to attach.
- 08 **Leila:** Keep it as your experience, then. A ⁴ _____ value needs measurement evidence. We shouldn't turn a comfort report into an unsupported volume-flow result.
- 09 **Tomas:** An earlier report used CFM. I assumed that was how fast the air moves. Does it actually describe something different?
- 10 **Leila:** It means ⁵ _____, a volume per unit time. It describes how much air volume moves, not the speed at one point by itself.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Tomas:** So a feeling of air movement at my seat doesn't give us a CFM figure. And liters per second would be another volume-flow unit.
- 12 **Leila:** Correct. ⁶ _____ is speed at a location, distinct from volume flow. Neither measurement is supplied in this report, although your location description is useful.
- 13 **Tomas:** Since we know which opening is the return, can we at least say the system is balanced? The drawing appears to show a complete arrangement.
- 14 **Leila:** Identifying the openings doesn't establish ⁷ _____. That involves technical distribution work and verification, not simply checking that supply and return labels exist.
- 15 **Tomas:** Someone suggested closing a grille to stop the discomfort. I don't want the service request to present that suggestion as the approved remedy.
- 16 **Leila:** It won't. No ⁸ _____ adjustment or grille change is prescribed by these records. The appropriate assessment needs to establish the condition and any required response.
- 17 **Tomas:** Please include that it's the seat below S3 that concerns me. Saying air-conditioning problem leaves the person reading it with very little to locate.
- 18 **Leila:** Agreed. The ⁹ _____ location matters to the report. We'll preserve where you experience discomfort without naming an equipment fault that hasn't been established.
- 19 **Tomas:** Read me the short version, please. I'll use that wording when the facilities manager asks why the request was raised.
- 20 **Leila:** Discomfort reported below supply S3; R2 labeled return; no airflow measurements supplied. No ¹⁰ _____ or cause is established. That keeps the complaint precise without inventing results.

ANSWERS / EXPLANATIONS: PAGE 90

1. supply air | 2. return air | 3. distribution report | 4. measured airflow | 5. cubic feet per minute | 6. Air velocity | 7. balancing | 8. damper | 9. occupied zone | 10. performance finding

Explain supply and return

F Say it

Complete the client summary. Identify each function and keep the missing measurement visible.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

supply / measurements / S3 / return

Technician: "S3 is labeled ¹ _____ air."

Client: "R2 is labeled ² _____ air."

Technician: "The discomfort is reported below ³ _____."

Client: "No airflow ⁴ _____ have been supplied."

ANSWERS / EXPLANATIONS: PAGE 90

1. supply | 2. return | 3. S3 | 4. measurements

Rehearse with a fixed script

1. Read turns 1-10. Use clear contrastive stress for supply into the room and return back toward the system.
2. Switch roles for turns 11-20. Distinguish CFM volume flow from air velocity and a comfort report from a measured result.
3. Complete and check the exchange. Keep the complaint below S3 and avoid converting the drawing labels into a balancing finding.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Five kilowatts of what?

A Read the situation

04

Compare like quantities, request missing equipment data, and distinguish refrigerant identification from approval to select or service a system.

Client Jules compares two fictional proposals with estimator Farah. Both show 5 kilowatts, but proposal A labels the value cooling capacity while proposal B labels it electrical input. Jules calls them equal-size systems. Full data sheets and equipment pairings are not supplied, and no selection is approved. Farah must explain why the matching numbers are not enough, request complete specifications, and keep refrigerant references tied to the actual equipment without recommending handling or substitution.

PEOPLE / Jules: Client | Farah: Estimator

ASK WHAT A FAMILIAR NUMBER MEASURES

Customers often compare the largest printed number in a proposal. Name the quantity before comparing the value. Explain what information is missing and why it matters, without turning a partial specification into a product recommendation or an efficiency claim.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does proposal A's 5 kilowatts describe?

- A Stored electrical energy
- C Cooling capacity

- B A refrigerant charge
- D Electrical input

2. What does proposal B's 5 kilowatts describe?

- A Confirmed equal cooling capacity
- C A maintenance interval

- B A measured room temperature
- D Electrical input

3. What conclusion is supported?

- A Both have identical efficiency
- C The numbers alone cannot establish equivalent systems

- B Proposal B is approved
- D Both systems have identical capacity

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

cooling capacity

Rate at which a system can remove heat under stated conditions.

IN USE compare cooling capacity

electrical input

Electrical power required by the equipment under stated conditions.

IN USE identify electrical input

kilowatt

Unit of power equal to one thousand watts.

IN USE state the rating in kilowatts

kilowatt-hour

Unit of energy representing one kilowatt over one hour.

IN USE distinguish a kilowatt-hour

rating condition

Defined condition under which a performance value is stated.

IN USE check the rating conditions

data sheet

Manufacturer document listing relevant product characteristics.

IN USE obtain the complete data sheet

equipment combination

Particular set of components proposed to work together.

IN USE verify the equipment combination

matched system

Component arrangement confirmed as suitable together by the relevant specifications.

IN USE identify the matched system

nominal capacity

Stated classification or approximate rating rather than every operating output.

IN USE distinguish nominal capacity

operating condition

Relevant environmental or system state during operation.

IN USE state the operating conditions

efficiency

Relationship between useful output and required input under defined conditions.

IN USE compare efficiency on a consistent basis

coefficient of performance

Ratio of useful heating or cooling output to input power on a defined basis.

IN USE interpret the coefficient of performance

Precision check

Both proposals state 5 kilowatts, but A describes cooling output and B electrical input. The unit can express either quantity. It does not follow that the systems have equal cooling capacities or equal efficiencies.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

refrigerant

Working fluid used to transfer heat in a refrigeration system.

IN USE identify the specified refrigerant

refrigerant designation

Identifier naming a particular refrigerant or blend.

IN USE preserve the refrigerant designation

refrigerant charge

Quantity of refrigerant specified or present in a system.

IN USE distinguish refrigerant charge

blend

Refrigerant made from more than one component substance.

IN USE identify a refrigerant blend

compatibility

Suitability of materials or components for use together.

IN USE verify compatibility

manufacturer specification

Product requirement or attribute stated by its manufacturer.

IN USE consult the manufacturer specification

superheat

Vapor temperature above its saturation reference at the same pressure; use dew for the stated blend.

IN USE report superheat in kelvin

selection basis

Information and criteria supporting an equipment choice.

IN USE establish the selection basis

like-for-like comparison

Comparison of matching quantities on a consistent basis.

IN USE make a like-for-like comparison

sensible heat ratio

Sensible cooling capacity divided by total cooling capacity on the same basis.

IN USE calculate the sensible heat ratio

subcooling

Liquid temperature below its saturation reference at the same pressure; use bubble for the stated blend.

IN USE distinguish subcooling from superheat

substitution review

Evaluation of a proposed replacement for a specified item.

IN USE request a substitution review

Precision check

The complete data sheets and component pairings are missing. A refrigerant designation must stay tied to the actual equipment specification; a familiar refrigerant name does not approve a substitution, determine compatibility, or authorize servicing.

Phrases you can actually use

C Notice the language / 1 of 2

ASK THE KEY QUESTION

"Five kilowatts of which quantity?"

READ PROPOSAL B

"B lists 5 kilowatts of electrical input."

REJECT THE FALSE EQUIVALENCE

"Matching numbers do not establish equal-size systems here."

INCLUDE THE COMBINATION

"The proposed component pairings also need to be identified."

READ PROPOSAL A

"A lists 5 kilowatts of cooling capacity."

NAME THE DISTINCTION

"Cooling output and electrical input are different."

ASK FOR COMPLETE DATA

"We need the full data sheets for both proposals."

KEEP CONDITIONS ATTACHED

"Compare ratings under the relevant stated conditions."

Of which quantity?

Forces a value to be connected to what it describes.

Both ... but

Acknowledges the shared number while exposing the important difference.

Under stated conditions

Prevents ratings from being treated as universal operating results.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID INVENTING EFFICIENCY

"We cannot calculate a supported efficiency comparison from these two entries."

KEEP REFRIGERANT INFORMATION SPECIFIC

"Use the refrigerant designation from the actual equipment specification."

KEEP CHARGE SEPARATE

"Refrigerant type and refrigerant quantity are different information."

STATE THE SELECTION LIMIT

"No equipment selection has been approved."

DISTINGUISH POWER AND ENERGY

"Kilowatts and kilowatt-hours are different units."

AVOID TREATING A NAME AS APPROVAL

"A refrigerant reference does not approve a substitution."

NAME THE REVIEW TASK

"We need a like-for-like comparison."

CLOSE WITH THE REQUEST

"Please supply complete ratings and equipment combinations for review."

Cannot calculate from these entries

Limits a conclusion to the information actually available.

Type versus quantity

Separates refrigerant identity from charge amount.

No selection approved

Keeps information gathering distinct from a purchase or installation decision.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which comparison is valid from the supplied entries?

- A** Both values prove identical energy consumption
- B** B must be more efficient than A
- C** A's value is cooling capacity; B's is electrical input
- D** Both values prove equal cooling output

2. What additional information is needed?

- A** Confirmation that both headlines still say five kilowatts
- B** A shared refrigerant name without the component information
- C** Full data sheets, rating conditions, and component pairings
- D** A ratio of A's output to B's input

3. Which conclusion exceeds the evidence?

- A** The systems have equal efficiency
- B** A is labeled cooling capacity
- C** B is labeled electrical input
- D** No selection is approved

4. What does a refrigerant designation provide?

- A** A complete compatibility assessment
- B** Permission for untrained servicing
- C** Identification to be checked against the actual equipment specification
- D** Automatic approval to substitute fluids

Answers and explanations: [page 91](#).

Five kilowatts of what?

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

cooling capacity / refrigerant designation / equipment combination / data sheet / Electrical input / substitution review / like-for-like comparison / Refrigerant charge / efficiency / selection basis

- 01 **Jules:** Farah, both proposals show five kilowatts. I put equal-size systems in my comparison sheet and moved straight to the two prices.
- 02 **Farah:** Let's check five kilowatts of what. Proposal A labels it ¹ _____. B uses a different quantity label even though its number and unit look the same.
- 03 **Jules:** B says electrical input. I treated kilowatts as though they always told us how much heat the system could remove from the room.
- 04 **Farah:** ² _____ is the power required by the equipment. Cooling capacity is heat-removal output under stated conditions, so these entries don't establish equal output.
- 05 **Jules:** Could I divide A's five by B's five and report a performance ratio of one? They're both power figures, after all.
- 06 **Farah:** Not for an ³ _____ comparison. You would be combining one system's output with another system's input, rather than corresponding quantities for each proposed arrangement.
- 07 **Jules:** Then asking each supplier to confirm five kilowatts again wouldn't fix it. We need both quantity labels and the missing values for each proposal.
- 08 **Farah:** Yes, request each full ⁴ _____, with the proposed components and rating conditions. A short headline doesn't establish performance across every operating situation.
- 09 **Jules:** I haven't received the indoor and outdoor pairings. Is a brochure for the outdoor unit enough if the installer chooses the indoor one later?
- 10 **Farah:** No. The ⁵ _____ matters to the stated performance. We need to know which components the proposal actually pairs, not assume any indoor unit gives the same result.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Jules:** One summary includes a refrigerant name. Could I use that as evidence that the two proposals are interchangeable replacements?
- 12 **Farah:** A ⁶ _____ identifies a fluid or blend, not an equivalent system. It must be checked against the actual equipment specification and doesn't settle all compatibility requirements.
- 13 **Jules:** I don't want the supplier to read our question as permission to use another fluid. We're asking for information, not changing the specified equipment.
- 14 **Farah:** Then keep any ⁷ _____ separate and explicitly pending. A familiar refrigerant name doesn't bypass that process or authorize refrigerant handling.
- 15 **Jules:** The sheet also has a refrigerant amount. Is that a different statement of capacity, or is it simply another product detail?
- 16 **Farah:** ⁸ _____ is the quantity of refrigerant specified or present. Keep the type and amount separate from cooling capacity and electrical input.
- 17 **Jules:** I see four different things now: fluid identity, fluid quantity, output power, and input power. The matching files only concern two of those.
- 18 **Farah:** Right. We need a ⁹ _____ using corresponding ratings and conditions. None of the missing values should be filled with the figure from the other proposal.
- 19 **Jules:** I'll remove equal-size and ask for complete data sheets, actual pairings, and both output and input values with their conditions.
- 20 **Farah:** That improves the ¹⁰ _____. It doesn't approve either option yet, but it gives the technical review the information needed to make a meaningful comparison.

ANSWERS / EXPLANATIONS: PAGE 92

1. cooling capacity | 2. Electrical input | 3. efficiency | 4. data sheet | 5. equipment combination | 6. refrigerant designation | 7. substitution review | 8. Refrigerant charge | 9. like-for-like comparison | 10. selection basis

Keep input and output apart

F Say it

Complete the proposal comparison. Preserve the different quantity labels and the missing approval.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

sheets / approved / input / capacity

Client: "A gives 5 kilowatts of cooling ¹ _____."

Estimator: "B gives 5 kilowatts of electrical ² _____."

Client: "We need complete data ³ _____ and pairings."

Estimator: "No selection is yet ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. capacity | 2. input | 3. sheets | 4. approved

Rehearse with a fixed script

1. Read turns 1-10. Attach cooling capacity to A and electrical input to B every time five kilowatts is mentioned.
2. Switch roles for turns 11-20. Keep refrigerant designation and charge separate, and make the missing pairing explicit.
3. Complete and check the comparison exchange. Request complete data without approving equipment or calculating across different proposals.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Yesterday wet, today dry, source still unknown

A Read the situation

Reconcile a historical report with a later observation and avoid treating an earlier maintenance concern as a proven cause.

Client Elena reported water near air handler AH-6 at 15:00 yesterday. Today's visit note states that the nearby floor is dry and the source was not identified. An earlier service log mentions a drain-pan concern but does not establish a link to the reported water. Coordinator Kai must preserve the chronology, distinguish each source, and summarize the unresolved issue. The exercise supplies records only; no equipment inspection or diagnostic procedure is requested.

PEOPLE / Elena: Client | Kai: Service coordinator

KEEP TWO TIMES VISIBLE IN ONE ACCOUNT

A later observation need not contradict an earlier report. Use time markers and source attribution rather than choosing one account and deleting the other. A prior concern can remain relevant history without becoming the confirmed cause of a new symptom.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What did Elena report yesterday?

- A A verified drain-pan failure
- C A dry floor throughout the day

- B A completed repair
- D Water near AH-6 at 15:00

2. What does today's visit note establish?

- A The earlier report was false
- C Every component was repaired

- B The drain pan caused the water
- D The nearby floor was dry and the source was not identified

3. What does the earlier drain-pan entry prove?

- A Only that a concern was recorded, not a causal link
- C That the concern has been repaired

- B That the pan caused yesterday's water
- D That the current system is fault-free

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

condensate

Liquid formed when water vapor condenses.

IN USE identify a condensate reference

drain pan

Component intended to collect relevant drainage or condensate in equipment.

IN USE record a drain-pan concern

condensate line

Pipe or tube associated with carrying condensate away.

IN USE identify the condensate line

moisture report

Account describing water or dampness at a location.

IN USE retain the moisture report

service log

Chronological record of service-related information.

IN USE consult the service log

visit note

Record of observations or actions during an attendance.

IN USE quote the visit note

historical entry

Record created before the current event.

IN USE distinguish a historical entry

chronology

Ordered sequence of events and observations.

IN USE preserve the chronology

reported water

Water described in an account before its cause is established.

IN USE locate the reported water

nearby floor

Floor surface close to the referenced equipment.

IN USE describe the nearby floor

source identification

Establishing where the reported water originated.

IN USE keep source identification open

causal link

Supported connection between a condition and its cause.

IN USE establish a causal link

Precision check

Yesterday at 15:00, Elena reported water near AH-6. Today, the floor is recorded as dry. Those observations concern different times; the later note does not disprove the earlier report or establish a repair.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

unresolved concern

Issue that has not been confirmed as corrected or closed.

IN USE retain an unresolved concern

dry observation

Record that the visible surface was dry at the stated time.

IN USE qualify the dry observation

intermittent symptom

Condition that may appear at some times and not others.

IN USE describe an intermittent symptom

recurrence history

Record of repeated appearances of a condition.

IN USE verify recurrence history

drainage path

Route intended for the relevant liquid to travel.

IN USE identify the drainage path

overflow

Liquid exceeding the capacity or boundary of its intended containment.

IN USE distinguish a reported overflow

leakage

Unintended escape of a fluid.

IN USE avoid assuming leakage source

condensation

Formation of liquid from vapor under relevant conditions.

IN USE distinguish condensation from a diagnosis

finding

Observation or conclusion established through the relevant assessment.

IN USE state the actual finding

repair status

Whether corrective work is proposed, underway, or completed.

IN USE verify repair status

evidence boundary

Limit of what the information supports.

IN USE preserve the evidence boundary

record reconciliation

Bringing separate accounts together without erasing their distinctions.

IN USE complete record reconciliation

Precision check

The earlier drain-pan concern is history, not a verified cause. Do not change near AH-6 to from AH-6, or concern to failure, without evidence. The source remains unidentified in the supplied visit note.

Phrases you can actually use

C Notice the language / 1 of 2

STATE YESTERDAY'S ACCOUNT

"The client reported water near AH-6 at 15:00 yesterday."

KEEP BOTH TIMES

"Those accounts refer to different observations at different times."

ATTRIBUTE THE HISTORY

"An earlier service-log entry mentions a drain-pan concern."

DISTINGUISH NEAR FROM FROM

"Near AH-6 gives a location, not a proven origin."

STATE TODAY'S OBSERVATION

"The visit note records the nearby floor as dry today."

NAME THE UNRESOLVED ISSUE

"The source was not identified."

AVOID CONVERTING HISTORY INTO CAUSE

"That entry does not establish a link to yesterday's water."

AVOID CHANGING CONCERN TO FAILURE

"The record says concern, not confirmed failure."

Yesterday versus today

Prevents observations at different times from being treated as a contradiction.

Near versus from

Distinguishes location from origin.

Mentions

Records what a document says without upgrading it to a verified diagnosis.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP REPAIR SEPARATE

"No completed repair is established by the dry-floor note."

USE THE EXACT EQUIPMENT REFERENCE

"Please retain AH-6 in the summary."

NAME THE THREE SOURCES

"Client report, visit note, and earlier log remain distinct."

GIVE A BALANCED SUMMARY

"Water reported yesterday; floor dry today; cause unresolved."

PRESERVE THE CLIENT REPORT

"The later observation does not cancel your account."

AVOID AN INVENTED RECURRENCE PATTERN

"We have not established how often this happens."

LIMIT THE FINDING

"We can report the surface condition without diagnosing the source."

CLOSE WITH ACCURATE STATUS

"The source question remains open for the relevant assessment."

Does not establish a link

Keeps a plausible history item from becoming a proven cause.

Recorded as

Attributes an observation to the supplied note.

Remains open

Preserves an unresolved matter without inventing an outcome.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which summary preserves chronology?

- A** Water from AH-6 yesterday at 15:00; floor dry today
- B** Water seen by the service team today; earlier pan concern
- C** Drain-pan failure yesterday; repair inferred from today's dry floor
- D** Water near AH-6 yesterday at 15:00; floor dry today; source unidentified

2. Which wording improperly asserts origin?

- A** The nearby floor was dry today
- B** The source was not identified
- C** Water came from AH-6
- D** Water was reported near AH-6

3. How should the drain-pan entry be handled?

- A** Delete it because the floor is dry
- B** Relabel it as a completed repair
- C** Retain it as history with no established causal link
- D** Treat it as the final diagnosis

4. What can the dry-floor note establish about repair?

- A** A drainage component was replaced
- B** No completed repair by itself
- C** Every possible leak was fixed
- D** The earlier concern was resolved

Answers and explanations: [page 93](#).

Yesterday wet, today dry, source still unknown

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

recurrence history / drain-pan concern / source identification / causal link / chronology / historical entry / record reconciliation / evidence boundary / visit note / repair status

- 01 **Elena:** Kai, yesterday I reported water near AH-6 at three. Today's note says dry floor. I don't want my report dropped because the visit happened later.
- 02 **Kai:** We can preserve the ¹ _____. Your account and today's note refer to different times, so a dry floor now does not cancel the water you reported yesterday.
- 03 **Elena:** Please keep yesterday and fifteen hundred attached to my account. Otherwise it could sound as though the technician saw water on today's visit.
- 04 **Kai:** I will. The ² _____ records today's nearby floor as dry and says the source was not identified. That is a separate observation with its own limits.
- 05 **Elena:** The earlier service log mentions a drain-pan concern. Someone suggested putting that as the cause so the record has an explanation.
- 06 **Kai:** We should retain the ³ _____ as history, but the entry does not establish a connection to yesterday's report. It is not enough to name the pan as the proven source.
- 07 **Elena:** I only know the water was near the air handler. Writing that it came from AH-6 would be more specific than what I observed.
- 08 **Kai:** Exactly. That is an ⁴ _____. Changing near AH-6 to from AH-6 would turn a location description into an unsupported statement about origin.
- 09 **Elena:** The old wording is concern, not failure. Could you retain that distinction rather than turning a historical note into a confirmed broken part?
- 10 **Kai:** It would change it. The ⁵ _____ says concern, not confirmed failure. We need to preserve that distinction even if the topic appears relevant to the current question.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

05

- 11 **Elena:** Then we'll have an unresolved source, but an accurate record. I don't want a neat explanation that neither the client report nor visit note establishes.
- 12 **Kai:** Then the ⁶ _____ stays open. We can report the facts and the earlier concern without claiming that the relationship between them has been verified.
- 13 **Elena:** Does dry today imply someone repaired it overnight? I haven't supplied any record of work between my report and the visit.
- 14 **Kai:** No. The ⁷ _____ cannot be inferred from the dry-floor observation. Nothing in these records establishes that a repair was completed between your report and the visit.
- 15 **Elena:** One older concern also doesn't establish a pattern of repeated water events. We shouldn't attach a frequency to the problem without the history.
- 16 **Kai:** Correct. A ⁸ _____ would need supporting information. We should not invent a frequency or treat every earlier note as another occurrence of the same water event.
- 17 **Elena:** Please combine the three records without losing their sources: my account, today's visit note, and the earlier drain-pan entry.
- 18 **Kai:** Client reported water near AH-6 at 15:00 yesterday. Today's floor was dry; source unidentified. Earlier log notes a drain-pan concern, with no established ⁹ _____ to the water.
- 19 **Elena:** That's the chronology I need. It neither dismisses yesterday's observation nor treats today's dry floor as proof that the source has been fixed.
- 20 **Kai:** I will use that ¹⁰ _____ in the handoff. The next assessment can address the source question with the original accounts intact, rather than starting from an assumed diagnosis.

ANSWERS / EXPLANATIONS: PAGE 94

1. chronology | 2. visit note | 3. drain-pan concern | 4. evidence boundary | 5. historical entry | 6. source identification | 7. repair status | 8. recurrence history | 9. causal link | 10. record reconciliation

Preserve the three records

F Say it

Complete the service summary. Keep yesterday, today, and the earlier concern distinct.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

dry / identified / concern / 15:00

Client: "Water was reported near AH-6 at ¹ _____ yesterday."

Coordinator: "Today the nearby floor was recorded as ² _____."

Client: "The earlier log mentions a drain-pan ³ _____."

Coordinator: "The source was not ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. 15:00 | 2. dry | 3. concern | 4. identified

Rehearse with a fixed script

1. Read turns 1-10. Stress yesterday at fifteen hundred, today dry, and earlier concern as three separate records.

2. Switch roles for turns 11-20. Keep near distinct from from, and dry distinct from repaired.

3. Complete and check the summary. Read the source status without treating historical concern as a confirmed cause.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

An alarm, a display value, and missing product data

A Read the situation

Report an alarm chronology accurately, distinguish cabinet and product information, and route a food-safety question to the responsible lead.

06

Cafe manager Noah discusses a refrigeration record with technician Imani. A cabinet display recorded 8 degrees Celsius at 06:10. Product temperatures are not supplied. An alarm-history entry and a scheduled-defrost entry have the same time, but event cause and duration are unconfirmed. Noah has already alerted the responsible food-safety lead under the site procedures; this discussion occurs alongside that escalation and does not delay protective action. He asks how the records relate to the food-safety question. The display value and coincident entries do not authorize food release or establish a harmless defrost event.

PEOPLE / Noah: Cafe manager | Imani: Refrigeration technician

GIVE THE RECORD, NOT AN IMPROVISED RELEASE DECISION

Operational pressure can turn a request for information into a request for reassurance. Identify exactly what the record contains and what is missing. Refer the product decision to the responsible food-safety lead while preserving the technical event as unresolved.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What was recorded at 06:10?

- A A confirmed harmless alarm cause
- B The end of a measured exposure period
- C A cabinet display value of 8 degrees Celsius
- D Verified temperatures for every food item

2. What do the matching timestamps establish?

- A Every food item remained safe
- B The two entries share a recorded time
- C Defrost definitely caused the alarm
- D The alarm lasted one minute

3. Who must decide the food-safety question?

- A The responsible food-safety lead under site procedures
- B The display value alone
- C The technician by assuming defrost
- D The customer waiting at the counter

Answers and explanations: page 95.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

refrigerated cabinet

Enclosed equipment used to hold products under cooled conditions.

IN USE identify the refrigerated cabinet

cabinet display

Interface showing a value or status from refrigeration equipment.

IN USE quote the cabinet display

product temperature

Temperature of the actual stored product.

IN USE distinguish product temperature

display value

Number shown by the equipment interface.

IN USE record the display value

alarm history

Log of recorded equipment alarm events.

IN USE review the alarm history

timestamp

Recorded date or time associated with an entry.

IN USE preserve the timestamp

defrost

Process intended to remove frost from relevant equipment surfaces.

IN USE identify the defrost entry

scheduled defrost

Defrost event listed in the control schedule or record.

IN USE distinguish scheduled defrost

event duration

Length of time an event continued.

IN USE establish event duration

coincident entries

Records appearing at the same stated time.

IN USE compare coincident entries

causation

Relationship in which one event produces another.

IN USE avoid assuming causation

temperature excursion

Departure from a defined temperature range or limit.

IN USE assess a recorded temperature excursion

06

Precision check

The record gives a display value of 8 degrees Celsius at 06:10. Product temperatures are absent. No safe holding period or release decision can be inferred from this isolated display value and timestamp.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

exposure history

Record of conditions experienced by the product over time.

IN USE establish exposure history

data gap

Missing information needed for an assessment.

IN USE identify the data gap

food-safety lead

Person assigned responsibility for the relevant food-safety decision.

IN USE notify the food-safety lead

site procedure

Established process applicable at the premises.

IN USE follow the site procedure

release decision

Authorization to make a product available for its intended use.

IN USE refer the release decision

product disposition

Decision about how affected products will be handled.

IN USE determine product disposition

alarm acknowledgment

Record that an alarm has been noticed or accepted by an operator.

IN USE distinguish alarm acknowledgment

alarm clearance

Change in alarm status that does not alone establish product safety.

IN USE distinguish alarm clearance

trend record

Series of values showing change over time.

IN USE review the trend record

sensor location

Position associated with a particular measurement device.

IN USE identify the sensor location

technical cause

Established equipment or process explanation for an event.

IN USE keep the technical cause open

escalation summary

Concise information passed to the responsible decision-maker.

IN USE give an escalation summary

Precision check

The alarm and scheduled-defrost entries coincide, but cause and duration remain unconfirmed. Do not call the event harmless or assume defrost explains it. Follow the site food-safety escalation process; the responsible lead decides product disposition.

Phrases you can actually use

C Notice the language / 1 of 2

QUOTE THE ACTUAL RECORD

"The cabinet display recorded 8 degrees Celsius at 06:10."

KEEP THE RECORD TYPE CLEAR

"That is a display value, not a verified food-temperature record."

AVOID A CAUSAL SHORTCUT

"The matching time does not establish that defrost caused the alarm."

REJECT PREMATURE REASSURANCE

"I cannot declare the food safe from these records."

NAME THE MISSING INFORMATION

"Product temperatures have not been supplied."

IDENTIFY THE COINCIDENT ENTRIES

"The alarm and scheduled-defrost entries share a timestamp."

PRESERVE DURATION UNCERTAINTY

"The event duration is unconfirmed."

NAME THE DECISION ROUTE

"The responsible food-safety lead must decide under the site procedures."

Recorded

Attributes the number to the equipment record without changing what it measures.

Same time versus because

Separates coincidence from a causal explanation.

Unconfirmed duration

Prevents a timestamp from being read as a complete event interval.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP TECHNICAL STATUS OPEN

"The cause of the event still needs confirmation."

PRESERVE THE DATA GAP

"The product exposure history is not established."

KEEP DISPOSITION SEPARATE

"Product disposition is separate from explaining the equipment entry."

READ BACK THE ESSENTIALS

"8 degrees at 06:10; product data absent; cause and duration unconfirmed."

AVOID EQUATING CLEARANCE WITH SAFETY

"Alarm clearance would not itself release the food."

ESCALATE ACCURATELY

"Please pass the display record and the missing data to the lead."

AVOID A GUESSED THRESHOLD

"We should not invent an allowable exposure period."

CLOSE WITHOUT RELEASING PRODUCT

"This technical discussion does not authorize food release."

Must decide under

Names both the responsible role and the governing site process.

Would not itself

Limits what an alarm-status change can establish.

Separate from

Keeps equipment assessment and product disposition as distinct questions.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

06

1. Which summary belongs in the escalation?

- A** Every product was measured at 8 degrees
- B** Alarm lasted only until 06:11
- C** Display 8 degrees Celsius at 06:10; product data absent; cause and duration unconfirmed
- D** Food definitely safe because defrost was scheduled

2. Why is defrost caused it unsupported?

- A** A scheduled entry proves the entire cycle completed normally
- B** A timestamp establishes both start and finish
- C** A defrost entry establishes every product's exposure history
- D** The shared timestamp does not establish causation

3. Who should receive the food-release question?

- A** Whoever recognizes the cabinet brand
- B** The person who wants service to resume fastest
- C** The alarm display as sole authority
- D** The responsible food-safety lead using site procedures

4. Which statement improperly fills a data gap?

- A** Product temperatures were not supplied
- B** The duration remains unconfirmed
- C** The food was warm for only one minute
- D** The display value was recorded at 06:10

Answers and explanations: [page 95](#).

An alarm, a display value, and missing product data

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

Alarm clearance / event duration / escalation summary / scheduled defrost / display value / Product disposition / product temperature / food-safety lead / exposure history / timestamp

- 01 **Noah:** Imani, we've alerted our food-safety lead and aren't treating this call as clearance to serve the affected food. Can you help explain the cabinet record?
- 02 **Imani:** Yes. The ¹ _____ is eight degrees Celsius at 06:10. It describes the equipment record, not a verified temperature for every stored item.
- 03 **Noah:** That's all I've been given for temperature: the cabinet display. There are no food readings in the information I'm sending to the lead.
- 04 **Imani:** Then ² _____ is missing. Keep that gap explicit; don't copy the display number into a field that would imply the actual food was measured.
- 05 **Noah:** The alarm history and the scheduled-defrost entry both say 06:10. Could I describe the alarm as caused by defrost?
- 06 **Imani:** The shared ³ _____ only tells us the entries coincide. It doesn't establish the cause, prove the event was harmless, or determine what happened to the products.
- 07 **Noah:** The owner asked how long it lasted. There's only that one recorded time in this extract, with no confirmed ending time.
- 08 **Imani:** Then ⁴ _____ is unconfirmed. A timestamp doesn't mean a one-minute event, and we shouldn't invent an interval to make the report look complete.
- 09 **Noah:** I'll keep the lead updated with those gaps. Our opening time is close, but the technical call isn't a substitute for the product decision.
- 10 **Imani:** Correct. The ⁵ _____ must decide under your site procedures. Follow that process while the equipment information is clarified; don't wait for a reassuring guess from the display.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Noah:** Please distinguish the scheduled item from the actual explanation. I want the handoff to say what the log contains without implying we've diagnosed the event.
- 12 **Imani:** It contains a ⁶ _____ entry at the same recorded time as the alarm. That association needs assessment; it isn't yet a confirmed causal explanation.
- 13 **Noah:** Suppose a later update says the alarm has disappeared. Would that alone answer whether the affected stock can be used?
- 14 **Imani:** No. ⁷ _____ describes alarm status, not the products' earlier conditions. It doesn't replace the responsible food-safety decision or establish an acceptable exposure history.
- 15 **Noah:** So I should leave the earlier product conditions unresolved, rather than assume that a cleared display would make the whole episode irrelevant.
- 16 **Imani:** Exactly. The ⁸ _____ isn't established by this extract. Keep the missing product data, duration, and cause visible alongside the known display reading.
- 17 **Noah:** My message says display eight Celsius at 06:10; no product readings supplied; alarm and defrost entries coincide; duration and cause unconfirmed.
- 18 **Imani:** That's a useful ⁹ _____. Send any relevant updates through the site process as they become available, keeping the product question separate from the equipment diagnosis.
- 19 **Noah:** The lead will handle the food decision under our procedures. I'll preserve the technical record and won't describe this call as approval to resume using the stock.
- 20 **Imani:** Right. ¹⁰ _____ belongs to that responsible process. We have clarified the supplied evidence, not declared the food safe or resolved the equipment event.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. display value | 2. product temperature | 3. timestamp | 4. event duration | 5. food-safety lead | 6. scheduled defrost | 7. Alarm clearance | 8. exposure history | 9. escalation summary | 10. Product disposition

Escalate the record without a safety claim

F Say it

Complete the handoff to the responsible lead. Preserve the display record and the missing product information.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

06:10 / supplied / lead / 8

Manager: "The display recorded ¹ _____ degrees Celsius."

Technician: "The timestamp was ² _____."

Manager: "Product temperatures were not ³ _____."

Technician: "The responsible food-safety ⁴ _____ must decide under site procedures."

ANSWERS / EXPLANATIONS: PAGE 96

1. 8 | 2. 06:10 | 3. supplied | 4. lead

Rehearse with a fixed script

1. Read turns 1-10. Keep the immediate site escalation in place while separating display temperature, product data, and event duration.
2. Switch roles for turns 11-20. Contrast a scheduled-defrost entry with established cause, and alarm clearance with product disposition.
3. Complete and check the handoff. Preserve the missing data and responsible lead; do not add a food-release claim.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Moving a grille is a coordinated change

A Read the situation

Acknowledge an appearance preference, explain unassessed consequences, and present two bounded choices without promising price or timing.

Office client Priya asks project coordinator Daniel to move a supply grille because it visually conflicts with a pendant light. The proposal has not assessed ductwork, ceiling restoration, service access, price, or timing. Priya may retain the current reviewed layout or request a coordinated change proposal. Daniel must explain the choice without treating a small visible move as a minor technical task or guaranteeing unchanged cost and completion.

PEOPLE / Priya: Office client | Daniel: Project coordinator

07

MAKE THE CHOICE CLEAR WITHOUT MINIMIZING THE WORK

A visible component can be connected to less visible work above the ceiling. Acknowledge the design preference, list the specific unassessed matters, and offer the stated choices. A request to assess a move is not approval to install it or an agreement to absorb its cost.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Why does Priya want the grille moved?

- A** A measured airflow failure is established
- B** A new layout has already been accepted
- C** A maintenance inspection ordered the move
- D** It visually conflicts with a pendant light

2. Which consequences have been assessed?

- A** None of the listed change effects
- B** Ductwork and price only
- C** Service access and timing only
- D** All effects with a fixed quote

3. What are the stated choices?

- A** Accept a guessed price or waive review
- B** Change the refrigerant or replace the system
- C** Retain the reviewed layout or request a coordinated change proposal
- D** Move it immediately or cancel the whole project

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

supply grille relocation

Proposed change to the position of a supply-air grille.

IN USE assess a supply grille relocation

pendant light

Light fitting suspended from the ceiling.

IN USE coordinate the pendant light

visual conflict

Appearance issue between elements in the same view.

IN USE describe the visual conflict

reflected ceiling plan

Drawing of ceiling elements represented as viewed in reflection from below.

IN USE review the reflected ceiling plan

ceiling void

Space above the visible ceiling and below the structure above.

IN USE coordinate the ceiling void

branch duct

Duct section branching from a larger distribution route.

IN USE review the branch duct

duct connection

Interface joining a duct to another component.

IN USE assess the duct connection

flexible duct

Air-conveying duct with a flexible construction.

IN USE identify the flexible duct

rigid duct

Air-conveying duct with a relatively fixed form.

IN USE distinguish rigid duct

service clearance

Space required for relevant maintenance or access.

IN USE preserve service clearance

ceiling restoration

Reinstatement of ceiling finishes after defined work.

IN USE assess ceiling restoration

access provision

Means included in the design for reaching services.

IN USE review the access provision

Precision check

The visible conflict concerns the grille and pendant light. No performance fault is established. Ductwork, restoration, access, price, and timing all need assessment before the proposed move can be treated as a defined change.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

reviewed layout

Arrangement that has undergone the stated project review.

IN USE retain the reviewed layout

change proposal

Defined suggestion for altering the existing arrangement.

IN USE request a change proposal

coordination review

Evaluation of the change across affected elements and trades.

IN USE arrange a coordination review

scope impact

Effect of a change on the work required.

IN USE identify the scope impact

cost impact

Effect of a change on price or expenditure.

IN USE assess the cost impact

schedule impact

Effect of a change on timing.

IN USE assess the schedule impact

design preference

Client choice concerning the desired arrangement or appearance.

IN USE record the design preference

technical feasibility

Whether the proposed arrangement can satisfy relevant technical conditions.

IN USE establish technical feasibility

revised layout

Updated arrangement proposed or accepted after a change.

IN USE review the revised layout

unassessed consequence

Effect not yet evaluated.

IN USE keep unassessed consequences visible

instruction to proceed

Direction authorizing agreed work to begin.

IN USE distinguish an instruction to proceed

decision record

Record of the option selected and its conditions.

IN USE confirm the decision record

Precision check

The two available decisions are to retain the reviewed layout or request a coordinated proposal. Choosing assessment does not approve a new location, authorize installation, or guarantee that the original cost and schedule will remain unchanged.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE PREFERENCE

"I understand why you want the grille clear of the pendant light."

AVOID MINIMIZING HIDDEN WORK

"The visible move may involve work above the ceiling."

SEPARATE APPEARANCE AND FAULT

"The concern is visual, not a confirmed performance defect."

STATE THE SECOND OPTION

"You can request a coordinated change proposal."

NAME THE REQUEST

"You are asking to relocate the supply grille."

IDENTIFY THE UNASSESSED ITEMS

"Ductwork, restoration, access, price, and timing still need assessment."

STATE THE FIRST OPTION

"You can retain the current reviewed layout."

KEEP THE CHOICE BOUNDED

"Those are the two decisions available at this stage."

May involve

Acknowledges possible hidden work without inventing a specific construction requirement.

Still need assessment

Keeps every listed consequence open.

Retain or request

Presents two concrete choices instead of an unbounded design task.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID A PRICE PROMISE

"I cannot confirm unchanged cost before review."

KEEP ACCESS IN SCOPE

"The proposal must account for service access."

PRESERVE TECHNICAL REVIEW

"The new position needs the relevant coordination."

READ BACK THE DECISION

"You are requesting assessment, not approving installation."

AVOID A TIMING PROMISE

"The schedule effect has not been established."

CLARIFY THE REQUEST STATUS

"Asking for a proposal does not authorize the relocation."

NAME RESTORATION EXPLICITLY

"Ceiling restoration must be included in the assessment."

CLOSE WITH THE NEXT DELIVERABLE

"We will bring back a defined proposal with its consequences."

Before review

Makes evidence a condition of a price or timing commitment.

Not a confirmed defect

Distinguishes an appearance preference from a performance finding.

With its consequences

Requires the proposal to show related effects, not merely a new dot on a plan.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which response explains the choice accurately?

- A** The move is already approved because the client prefers it
- B** Only the grille position matters; access can be ignored
- C** Retain the reviewed layout or request a coordinated proposal with impacts assessed
- D** The move is free because it is visually small

2. Which item must not disappear from review?

- A** A promised unchanged date
- B** Service access
- C** A guessed refrigerant change
- D** An invented system failure

3. What does requesting a proposal authorize here?

- A** Assessment of the change, not its installation
- B** Immediate relocation
- C** Automatic acceptance of any price
- D** Waiving ceiling restoration

4. Which phrase overstates certainty?

- A** We need to assess the ductwork
- B** The conflict is visual
- C** You can retain the current layout
- D** We can move it with no cost or timing effect

[Answers and explanations: page 97.](#)

Moving a grille is a coordinated change

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

reviewed layout / change proposal / access provision / schedule impact / cost impact / ductwork / Ceiling restoration / visual conflict / decision record / design preference

07

- 01 **Priya:** Daniel, the supply grille looks awkward beside the pendant light. I'd like it moved. Is the small distance on the ceiling a small change to the work?
- 02 **Daniel:** It is a ¹ _____ we can record, but the visible size of the move does not tell us the amount of work involved above the ceiling.
- 03 **Priya:** Yes, it's an appearance issue. I'm not reporting poor cooling or saying the grille has failed; I want the light and grille to sit better together.
- 04 **Daniel:** Understood. We should call it a ² _____, not an established performance defect. That keeps the reason for the request accurate when it reaches the design team.
- 05 **Priya:** What would need assessment beyond marking a new position on the ceiling plan? I hadn't thought about the connection above it.
- 06 **Daniel:** The ³ _____ has not been assessed, nor have ceiling restoration, service access, price, or timing. A revised drawing must be supported by review of those related effects.
- 07 **Priya:** Would restoring the old ceiling opening be part of that review too? I don't want a price that leaves the finishing work unexplained.
- 08 **Daniel:** Yes. ⁴ _____ must be assessed rather than treated as automatically included or unnecessary. We should not give you a price for only part of the actual change.
- 09 **Priya:** Can we also check service access? Moving something to a neater position wouldn't help if it makes the relevant services hard to reach.
- 10 **Daniel:** Correct. The ⁵ _____ needs review with the other requirements. Appearance is important, but it does not by itself establish whether a revised arrangement is technically acceptable.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Priya:** What decision can I make now? I don't have the information to design the duct route or choose a technically acceptable new position myself.
- 12 **Daniel:** You can retain the current ⁶ _____, or request a coordinated change proposal. Those are the two options at this stage; neither requires you to specify a duct route.
- 13 **Priya:** I'd like a proposal. Please don't treat that request as acceptance of an unknown extra cost or as permission to start the move.
- 14 **Daniel:** It does not. A ⁷ _____ will define the alternative and its consequences for review. Requesting it is separate from accepting the resulting work and commercial terms.
- 15 **Priya:** Can you keep the original finish date? The office move-in depends on it, though I realize the schedule effect hasn't been assessed yet.
- 16 **Daniel:** I cannot confirm that yet. The ⁸ _____ is one of the open questions, along with cost. Your timing concern should be recorded without being mistaken for a promise that no effect exists.
- 17 **Priya:** Then please record timing as a concern, not as your guarantee. And avoid calling the move free or minor before the full scope is known.
- 18 **Daniel:** Agreed. The ⁹ _____ will need a defined scope. We will include ductwork, access, restoration, and timing in the coordinated review rather than price only the visible grille.
- 19 **Priya:** That's my decision: assess a coordinated proposal and return the consequences for review. I'm not instructing the team to relocate the grille now.
- 20 **Daniel:** That is the ¹⁰ _____: coordinated proposal requested; relocation not authorized. The current reviewed arrangement is not replaced by an assumed new position during this conversation.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. design preference | 2. visual conflict | 3. ductwork | 4. Ceiling restoration | 5. access provision | 6. reviewed layout | 7. change proposal | 8. schedule impact | 9. cost impact | 10. decision record

Offer two clear paths

F Say it

Complete the choice discussion. Keep the existing layout and the request for assessment separate from installation approval.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

proposal / assessment / layout / relocation

Coordinator: "You can retain the reviewed ¹ _____."

Client: "Or request a coordinated change ² _____."

Coordinator: "Cost and timing still need ³ _____."

Client: "This request does not authorize ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. layout | 2. proposal | 3. assessment | 4. relocation

Rehearse with a fixed script

1. Read turns 1-10. Name the visual issue and each unassessed effect: ductwork, restoration, access, cost, and timing.

2. Switch roles for turns 11-20. Stress retain the layout or request a proposal, with no approval to relocate.

3. Complete and check the choice exchange. Keep a request for assessment separate from agreed work and an unchanged finish date.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Report received, seasonal check still open

A Read the situation

Separate document receipt from completed verification and distinguish assigned correction work from an unassigned scheduling responsibility.

Client Ben receives a commissioning report for a fictional office system. It lists a deferred seasonal check, and the maintenance sheet names the wrong model. Coordinator Nia accepts responsibility for document correction and will update Ben on Friday. The seasonal-check date is not booked, and responsibility for scheduling it has not been confirmed. The handover must preserve that ownership gap. No overall performance conclusion, completed seasonal result, or change to warranty terms is supplied.

PEOPLE / Ben: Client | Nia: Coordinator

08

DO NOT LET ONE NAMED OWNER HIDE A DIFFERENT UNASSIGNED TASK

A coordinator can own a document correction without owning every remaining technical action. A useful handover names the received records, the unresolved checks, and who is responsible for each. State an ownership gap directly rather than letting a polite promise to update imply it is settled.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does the commissioning report list?

- A A completed seasonal result
- C No outstanding verification

- B A confirmed Friday test booking
- D A deferred seasonal check

2. What responsibility does Nia accept?

- A An automatic warranty extension
- C Confirmed ownership of scheduling every check

- B Document correction and a Friday update
- D A guaranteed system-performance result

3. What remains unresolved about the seasonal check?

- A Whether any report was received
- C Both the date and the scheduling owner

- B Whether Nia agreed to correct the sheet
- D Only the spelling of Friday

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

commissioning

Process of verifying and documenting system performance against relevant project requirements.

IN USE distinguish commissioning from document delivery

commissioning report

Record of commissioning activities, results, and outstanding matters.

IN USE receive the commissioning report

deferred check

Verification postponed until a later appropriate time or condition.

IN USE record the deferred check

seasonal check

Verification associated with relevant seasonal operating conditions.

IN USE schedule the seasonal check

maintenance sheet

Document describing relevant maintenance information for equipment.

IN USE correct the maintenance sheet

model-specific guidance

Information applicable to the actual identified product model.

IN USE obtain model-specific guidance

Precision check

The report has been received, but the seasonal check is deferred. The wrong-model maintenance sheet needs correction. Neither delivery nor a corrected document would itself establish a completed seasonal result or overall system performance.

document mismatch

Difference between supplied information and the relevant equipment.

IN USE identify the document mismatch

correction owner

Person responsible for resolving a record error.

IN USE name the correction owner

scheduling owner

Person responsible for arranging the required attendance or check.

IN USE confirm the scheduling owner

ownership gap

Task for which responsibility has not been confirmed.

IN USE preserve the ownership gap

unbooked check

Required verification with no confirmed appointment.

IN USE retain the unbooked check

update commitment

Promise to provide further information at a stated time.

IN USE honor the update commitment

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

outstanding verification

Check not yet completed or accepted.

IN USE track outstanding verification

handover register

Record of delivered information and remaining handover matters.

IN USE update the handover register

operating record

Documented information about actual operation.

IN USE retain the operating record

performance result

Established outcome from relevant measurement or verification.

IN USE distinguish a performance result

acceptance criterion

Requirement used to judge whether an outcome meets the agreed standard.

IN USE identify the acceptance criterion

test condition

Defined state under which a verification is carried out.

IN USE record the test condition

deferral reason

Explanation for postponing an activity.

IN USE preserve the deferral reason

closeout status

Current position of remaining completion matters.

IN USE state the closeout status

maintenance requirement

Relevant action or condition specified for upkeep.

IN USE verify the maintenance requirement

warranty terms

Actual conditions and coverage stated in the relevant warranty.

IN USE preserve warranty terms

document receipt

Confirmation that information has been delivered.

IN USE acknowledge document receipt

responsibility readback

Repetition of assigned and unassigned tasks for clarity.

IN USE give a responsibility readback

Precision check

Nia owns document correction and promises a Friday update. The seasonal check has no booked date and no confirmed scheduling owner. Preserve both gaps instead of silently assigning the check to Nia or treating Friday as its appointment.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE REPORT

"You have received the commissioning report."

IDENTIFY THE DOCUMENT ERROR

"The maintenance sheet names the wrong model."

COMMIT TO AN UPDATE

"I will update you on Friday."

STATE THE BOOKING GAP

"The seasonal-check date is not booked."

NAME THE EXCEPTION

"The report lists a deferred seasonal check."

ACCEPT A SPECIFIC TASK

"I will take responsibility for document correction."

KEEP THE DATE LIMITED

"Friday is an update date, not a booked seasonal check."

STATE THE OWNERSHIP GAP

"Responsibility for arranging that check is not confirmed."

08

Lists a deferred check

Describes a recorded exception rather than a completed result.

Will update

Commits to communication without guaranteeing an unresolved outcome.

Responsibility not confirmed

Makes an ownership gap explicit.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID IMPLIED OWNERSHIP

"My document task does not settle who schedules the check."

SEPARATE RECEIPT AND VERIFICATION

"Receiving the report does not complete the deferred check."

KEEP WARRANTY SEPARATE

"This discussion does not change the actual warranty terms."

READ BACK RESPONSIBILITIES

"Nia owns correction; seasonal scheduling ownership remains open."

PRESERVE MODEL SPECIFICITY

"The replacement sheet must match the actual model."

AVOID AN OVERALL ASSURANCE

"We have no overall performance conclusion to add here."

NAME THE NEXT COORDINATION ISSUE

"We still need a confirmed scheduling owner and date."

CLOSE WITH ACCURATE STATUS

"Documents are received, corrections and deferred verification remain visible."

08

Does not settle

Prevents one accepted task from silently resolving another.

Must match

Connects the maintenance document to the relevant equipment model.

Remain visible

Keeps unfinished items in the handover instead of hiding them in a general completion claim.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which summary preserves all responsibilities?

- A** The client automatically owns the seasonal check
- B** Nia corrects documents and updates Friday; seasonal-check owner and date remain open
- C** Nia has booked every check for Friday
- D** Report receipt proves full performance

2. What does Friday identify?

- A** Completion of every open item
- B** Nia's promised update
- C** A confirmed seasonal test date
- D** A warranty start date supplied by this case

3. Which inference is unsupported?

- A** Scheduling ownership is unconfirmed
- B** Correcting the maintenance sheet completes the seasonal check
- C** The sheet names the wrong model
- D** The report was received

4. What should the handover keep open?

- A** Only a date, because Nia's document task assigns her the seasonal check
- B** A Friday test appointment inferred from the update commitment
- C** Correct model guidance, deferred verification, scheduling owner, and date
- D** Only the leaflet, because receiving the report completes verification

Answers and explanations: page 99.

Report received, seasonal check still open

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

model-specific guidance / update commitment / warranty terms / document mismatch / ownership gap / handover register / responsibility readback / deferred check / performance result / scheduling owner

08

- 01 **Ben:** Nia, the commissioning report arrived, but it lists a deferred seasonal check. The maintenance sheet also names a different model from our actual system.
- 02 **Nia:** Those are two distinct handover matters. I accept the ¹ _____ on the maintenance sheet and will take responsibility for correcting that information.
- 03 **Ben:** Thanks for taking the document correction. I want to separate that from the outstanding check rather than mark everything complete on receipt of the report.
- 04 **Nia:** No. The ² _____ is still outstanding. The report records its status; delivery of that report does not carry out the verification or supply its result.
- 05 **Ben:** Who is arranging the seasonal check? I also can't find a booked date, so both responsibility and scheduling seem unresolved.
- 06 **Nia:** The ³ _____ has not been confirmed, and no date is booked. I should not imply that my document-correction task resolves either of those questions.
- 07 **Ben:** I nearly assigned that to you because you're handling the wrong sheet. That would add a responsibility you haven't actually accepted.
- 08 **Nia:** Exactly. We need to preserve the ⁴ _____ for the seasonal check. It needs an explicit assignment, not an assumption based on who is speaking at handover.
- 09 **Ben:** What date can I put down for the next update? I need a communication commitment even while the technical appointment remains unbooked.
- 10 **Nia:** I will correct the document information and make an ⁵ _____ for Friday. That gives you a communication date without presenting an unbooked seasonal check as scheduled.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Ben:** Please match the corrected sheet to the actual model. A similar product leaflet would still leave the maintenance information uncertain.
- 12 **Nia:** Agreed. We need ⁶ _____ for the actual equipment. Correcting the sheet is a defined records task; it is not evidence that the seasonal verification has occurred.
- 13 **Ben:** Would report received support a statement that the system has passed every operating condition? The deferred check suggests that would be too broad.
- 14 **Nia:** No. We have no overall ⁷ _____ to add from these facts. The deferred item remains part of the record, and we should not invent a successful result for it.
- 15 **Ben:** I also want the summary to preserve the actual warranty terms, without implying the open check creates a new warranty promise.
- 16 **Nia:** Correct. The ⁸ _____ are not changed by this conversation. Neither an update date nor an outstanding check creates a new warranty promise here.
- 17 **Ben:** Let me check the responsibilities: you own document correction and the Friday update. The seasonal-check owner and date are both still unconfirmed.
- 18 **Nia:** That ⁹ _____ is accurate. It makes the commitment and both remaining scheduling questions visible instead of assigning the whole handover to one person by implication.
- 19 **Ben:** I'll acknowledge receipt and keep the unresolved items separate. Friday goes in the update column, not the seasonal-test appointment column.
- 20 **Nia:** Thank you. The ¹⁰ _____ should preserve the report receipt, wrong-model correction, deferred check, and open scheduling responsibility and date. That is a clear close to today's conversation, with the unfinished work still visible.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. document mismatch | 2. deferred check | 3. scheduling owner | 4. ownership gap | 5. update commitment | 6. model-specific guidance | 7. performance result | 8. warranty terms | 9. responsibility readback | 10. handover register

Assign only the responsibility actually accepted

F Say it

Complete the handover readback. Keep the document owner, update date, and unassigned seasonal-check task distinct.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

Nia / confirmed / Friday / unbooked

Client: "1 _____ owns document correction."

Coordinator: "The update is promised for 2 _____."

Client: "The seasonal check is still 3 _____."

Coordinator: "Its scheduling owner is not yet 4 _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. Nia | 2. Friday | 3. unbooked | 4. confirmed

Rehearse with a fixed script

1. Read turns 1-10. Separate document correction from seasonal-check scheduling, stating both the owner gap and the missing date.
2. Switch roles for turns 11-20. Keep Friday as an update and report receipt separate from a completed performance check.
3. Complete and check the responsibility exchange. Name Nia only for the task she accepted; do not silently assign the seasonal check.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Total cooling hides the split

E More workplace conversations / 1-10

Compare sensible and latent cooling separately instead of relying on one total rating.

Fictional comparison on a consistent net-capacity basis at the same stated operating conditions: room load is 10 kW sensible plus 2 kW latent. Option A supplies 12 kW total, including 9 kW sensible; B supplies 13 kW total, including 10.4 kW sensible. Total = sensible + latent; sensible heat ratio (SHR) = sensible / total. Actual equipment selection and installation are not approved.

PEOPLE / Mina: Facilities manager | Theo: HVAC estimator

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

load profile / sensible heat ratio / latent load / comparison basis / sensible load / total capacity

- 01 **Mina:** Option A says twelve kilowatts, exactly the room's total load. Can I mark it as meeting the requirement before we compare the price?
- 02 **Theo:** We need the split. The room's ¹ _____ is ten kilowatts, while A's sensible capacity is nine. Matching the totals hides a one-kilowatt shortfall.
- 03 **Mina:** And the other two kilowatts in the room load concern moisture removal. They aren't another two degrees of temperature reduction.
- 04 **Theo:** Correct. That's the ² _____. Sensible cooling concerns temperature change; latent cooling here concerns condensation of water vapor removed from the air.
- 05 **Mina:** If A has twelve total and nine sensible, its remaining capacity is three latent. That's one more than the latent load.
- 06 **Theo:** Yes, but the ³ _____ doesn't make the two parts interchangeable. The extra latent capacity doesn't establish ten kilowatts of sensible cooling at this condition.
- 07 **Mina:** The worksheet also lists SHR. For A, nine divided by twelve gives zero point seven five. Does that mean seventy-five percent efficiency?
- 08 **Theo:** No. The ⁴ _____ describes the sensible share of cooling, not electrical efficiency. We haven't been given input power for an efficiency calculation.
- 09 **Mina:** B gives thirteen total and ten point four sensible. Subtracting leaves two point six latent, so both supplied load components are covered numerically.
- 10 **Theo:** Yes, on the stated ⁵ _____. B's SHR is ten point four divided by thirteen, or zero point eight.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Mina:** A bigger sensible ratio doesn't automatically mean a better unit, then. The right split depends on what the space needs, not the highest SHR.
- 12 **Theo:** Exactly. A room with a different ⁶ _____ could need a different balance. We compare the relevant components and conditions, not a ratio as a universal quality score.
- 13 **Mina:** Can the brochure's headline capacity replace these values? It may have been rated at a different outdoor temperature and entering-air condition.
- 14 **Theo:** Not without checking the conditions. Use the performance data for the comparison being made, including airflow and the equipment combination, rather than swapping in a favorable headline.
- 15 **Mina:** The case says net capacity. If another sheet says gross coil capacity, should I use it directly because both are still in kilowatts?
- 16 **Theo:** No. Gross and net values can have different treatment of fan heat. Reconcile the stated basis before comparing them; identical units don't guarantee equivalent boundaries.
- 17 **Mina:** Then B clears this worksheet's sensible and latent checks, but we still haven't selected equipment for the actual building.
- 18 **Theo:** Right. Part-load behavior, controls, conditions, and the rest of the design assessment remain relevant. This arithmetic is not approval to purchase or install a system.
- 19 **Mina:** I'll record A as nine sensible and three latent, with a one-kilowatt sensible shortfall. B is ten point four sensible and two point six latent.
- 20 **Theo:** Include the load split, rating conditions, and net-capacity basis. That explains why a twelve-kilowatt total alone was not enough to settle the comparison.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. sensible load | 2. latent load | 3. total capacity | 4. sensible heat ratio | 5. comparison basis | 6. load profile

Check the components, not just the total

F Say it in a new situation

Fictional like-for-like net-capacity comparison: load 6 kW sensible and 2 kW latent. Option C: 8 kW total, 6.4 kW sensible. Option D: 9 kW total, 6.3 kW sensible. Use latent = total - sensible and SHR = sensible / total. Compare only these stated figures.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

0.4 / 1.6 / 0.7 / D

Manager: C's latent capacity is ¹ _____ kW.

Estimator: C therefore has a latent shortfall of ² _____ kW.

Manager: Option ³ _____ covers both supplied component loads numerically.

Estimator: D's sensible heat ratio is ⁴ _____ .

ANSWERS / EXPLANATIONS: PAGE 101

1. 1.6 | 2. 0.4 | 3. D | 4. 0.7

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

Circulation is not all outdoor air

E More workplace conversations / 1-10

Explain an air-balance report using consistent volume, time, and outdoor-air quantities.

Fictional steady-flow room report: volume 300 cubic meters; total supply 600 cubic meters per hour, comprising 150 outdoor and 450 recirculated. Assume these flows serve this room on the same volume basis. Ignore other air paths for this arithmetic only. Air changes per hour = hourly flow / room volume. No ventilation requirement, filtration effectiveness, or health assurance is supplied.

PEOPLE / Omar: Office coordinator | Leah: Ventilation specialist

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

air changes per hour / time basis / outdoor-air flow / outdoor-air fraction / total supply / air mixing

10

- 01 **Omar:** The report gives six hundred cubic meters an hour for a three-hundred-cubic-meter room. I wrote two complete fresh-air replacements every hour. Is that accurate?
- 02 **Leah:** Not as written. Dividing the ¹ _____ by room volume gives two air changes per hour, but the supply includes recirculated air.
- 03 **Omar:** Only one hundred fifty of the six hundred comes from outside. I shouldn't call all six hundred fresh outdoor air just because it enters the room.
- 04 **Leah:** Correct. The ² _____ is one hundred fifty cubic meters per hour. The other four hundred fifty is recirculated within the stated arrangement.
- 05 **Omar:** Then the outdoor-air calculation is one hundred fifty divided by three hundred, or zero point five. I need to label both results.
- 06 **Leah:** Yes: two total-supply and zero point five outdoor-air ³ _____. The shared unit doesn't mean the two figures represent the same air source.
- 07 **Omar:** If I want the percentage of supply that is outdoor air, I divide one hundred fifty by six hundred rather than by room volume.
- 08 **Leah:** Exactly. The ⁴ _____ is one quarter, or twenty-five percent. Don't confuse that fraction with zero point five air changes per hour.
- 09 **Omar:** The equipment summary uses liters per second instead. Six hundred cubic meters an hour would be about one hundred sixty-six point seven liters per second?
- 10 **Leah:** Yes. Keep the ⁵ _____ straight: multiply by one thousand, then divide by thirty-six hundred. Outdoor flow is about forty-one point seven liters per second.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Omar:** Does two total air changes mean every bit of the original room air has gone after half an hour? The word replacement makes it sound that way.
- 12 **Leah:** No. Actual ⁶ _____ and distribution matter, and much of this supply is recirculated. One room-volume of delivery doesn't sweep out every original molecule.
- 13 **Omar:** Can the recirculated portion still contribute to particle removal if the system has appropriate filtration? I don't want to say it has no possible benefit.
- 14 **Leah:** It can, but this report gives no filtration-effectiveness evidence for a clean-air calculation. Don't count all recirculation as clean air automatically, or call it outdoor air.
- 15 **Omar:** So we can't combine the two total-supply changes and the half outdoor change and advertise two and a half. The outdoor portion is already inside the total.
- 16 **Leah:** Exactly. That would count the outdoor flow twice. Six hundred already equals one hundred fifty plus four hundred fifty under the case's stated assumptions.
- 17 **Omar:** And none of these calculations says the room meets its actual ventilation requirement or makes the air safe for a particular use.
- 18 **Leah:** Correct. Occupancy, the applicable design requirements, distribution, filtration, and other conditions need assessment. This record alone isn't a compliance or health determination.
- 19 **Omar:** I'll replace fresh-air replacements with two total-supply ACH, including zero point five outdoor-air ACH, and state the twenty-five-percent outdoor fraction.
- 20 **Leah:** Keep the room volume and original flow units alongside those figures. That lets the reader check both the arithmetic and exactly which air quantity each result describes.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. total supply | 2. outdoor-air flow | 3. air changes per hour | 4. outdoor-air fraction | 5. time basis | 6. air mixing

Keep the two air-change rates apart

F Say it in a new situation

Fictional room volume 240 cubic meters; total supply 720 cubic meters per hour, including 180 outdoor and 540 recirculated. Same volume basis; no other air paths in this calculation. No adequacy or safety judgment is requested.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

50 / 0.75 / three / 25

10

Coordinator: Total-supply air changes per hour are ¹ _____.

Specialist: Outdoor-air changes per hour are ² _____.

Coordinator: Outdoor air forms ³ _____ percent of the supply.

Specialist: Outdoor flow is ⁴ _____ liters per second.

ANSWERS / EXPLANATIONS: PAGE 102

1. three | 2. 0.75 | 3. 25 | 4. 50

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

Use the right saturation reference

E More workplace conversations / 1-10

Explain superheat and subcooling in a completed report without using the wrong blend reference.

Fictional completed report for a refrigerant blend: at the vapor-reading pressure, bubble point 0 C, dew point 5 C, vapor temperature 12 C; at the liquid-reading pressure, bubble point 40 C, dew point 45 C, liquid temperature 35 C. Each reference matches its reading's pressure. Superheat = vapor minus dew; subcooling = bubble minus liquid. No targets, diagnosis, or servicing instructions are supplied.

PEOPLE / Sara: Service coordinator | Dev: Refrigeration technician

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

superheat / subcooling / temperature glide / matching pressure / bubble point / dew point

11

- 01 **Sara:** Dev, the completed report gives two saturation temperatures for each pressure. I used zero for the vapor calculation and got twelve degrees of superheat.
- 02 **Dev:** For this blend's vapor reading, use the ¹ _____ of five degrees Celsius. Zero is the bubble point at that pressure, not the correct reference for this superheat calculation.
- 03 **Sara:** Then twelve minus five gives seven. I need to correct the derived result, but preserve twelve as the original vapor temperature.
- 04 **Dev:** Yes. The ² _____ is seven kelvin, a temperature difference. We aren't changing the measured temperature or saying the vapor is at seven degrees Celsius.
- 05 **Sara:** The liquid-side entry gives forty bubble and forty-five dew, with liquid at thirty-five. Which reference belongs with that calculation?
- 06 **Dev:** Use the ³ _____ of forty degrees Celsius. The liquid calculation is referenced to saturated liquid at its corresponding pressure, not to the dew value.
- 07 **Sara:** Forty minus thirty-five is five. Using forty-five would give ten and overstate the result by another five.
- 08 **Dev:** Correct. The ⁴ _____ here is five kelvin. Don't swap the reference merely because the larger subtraction looks like a more substantial result.
- 09 **Sara:** Why aren't the bubble and dew points identical? I expected one saturation temperature, as though the whole change of state happened at a single value.
- 10 **Dev:** The blend has ⁵ _____. Its bubble and dew temperatures differ at a given pressure. Keep the two reference labels instead of silently averaging them.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Sara:** Both pairs in this example differ by five. Can I use the vapor-side dew point with the liquid temperature just because both numbers appear in one report?
- 12 **Dev:** No. Each comparison needs the ⁶ _____ reference for that reading and location. The report already supplies those pairings; crossing between them would invalidate the result.
- 13 **Sara:** Is seven kelvin a different-sized interval from seven Celsius degrees? I can see how the report's units might confuse someone reading quickly.
- 14 **Dev:** Those intervals are the same size. Kelvin is useful for stating a temperature difference here; it doesn't turn the seven-kelvin difference into an absolute temperature of seven kelvin.
- 15 **Sara:** Do seven and five tell us the system has the correct charge or that an expansion device needs adjustment? The summary asks for the service outcome.
- 16 **Dev:** Not from these figures alone. No applicable targets or full assessment are supplied. Correct arithmetic doesn't establish system condition, refrigerant quantity, or a remedy.
- 17 **Sara:** Then the handoff should distinguish corrected calculations from completed servicing. Nobody has been asked to obtain new readings or change a setting in this exercise.
- 18 **Dev:** Exactly. We're reviewing a completed record. Actual testing, refrigerant work, and adjustments require the relevant qualified procedures and authorization, not a language-book calculation.
- 19 **Sara:** I'll report vapor twelve Celsius against dew five: superheat seven kelvin. Liquid thirty-five against bubble forty: subcooling five kelvin, with the original location references retained.
- 20 **Dev:** That's clear. Keep the underlying readings, pressure-specific references, and unresolved assessment status visible so the corrected summary doesn't become an unsupported all-clear.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. dew point | 2. superheat | 3. bubble point | 4. subcooling | 5. temperature glide | 6. matching pressure

Recalculate without changing the readings

F Say it in a new situation

Completed fictional blend record: vapor temperature 11 C with matching dew 4 C and bubble -1 C; liquid temperature 31 C with matching bubble 37 C and dew 42 C. Use the correct reference for each calculation. No operating target or diagnosis is supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

bubble / seven / six / dew

11

Coordinator: The vapor calculation uses the ¹ _____ point.

Technician: The superheat is ² _____ kelvin.

Coordinator: The liquid calculation uses the ³ _____ point.

Technician: The subcooling is ⁴ _____ kelvin.

ANSWERS / EXPLANATIONS: PAGE 103

1. dew | 2. seven | 3. bubble | 4. six

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Clarifying comfort complaints and thermostat readings

A Briefing checks / page 4

1. What does 22 degrees Celsius represent?

B. The thermostat setpoint

The brief identifies the number as the target setting rather than a measured room condition.

2. When do staff report feeling warm?

C. After 14:00

The supplied pattern is an afternoon report beginning after 14:00.

3. What is established about the equipment?

B. No cause or fault has been established

The complaint and setpoint do not establish a cause or equipment condition.

D Language checks / page 9

1. Which wording accurately uses the number?

A. The thermostat setpoint is 22 degrees Celsius

The value identifies a target setting, not any of the proposed measurements.

2. Which summary preserves the complaint?

B. Staff feel warm in the meeting room after 14:00; ambient reading not supplied

The summary keeps attribution, place, period, and missing measurement intact.

3. What should Ben avoid inferring?

C. A particular equipment fault

No diagnostic evidence establishes a component failure in the supplied facts.

4. Which question best clarifies the displayed value?

A. Does the record label 22 as the target or the measured room temperature?

The bounded question identifies the value using existing information without assuming a fault.

01 Conversation answers

Clarifying comfort complaints and thermostat readings

E Dialogue gaps

Pages 10-11

- 1. setpoint.** Turn 2. Setpoint identifies the target setting, which must not be mistaken for a room measurement.
- 2. ambient temperature.** Turn 4. Ambient temperature is the surrounding-air condition, for which no confirmed reading is supplied.
- 3. afternoon pattern.** Turn 6. Afternoon pattern preserves the reported period after 14:00 without inventing continuous discomfort or exact onset.
- 4. occupant feedback.** Turn 8. Occupant feedback records the staff experience alongside the target rather than treating one as a contradiction.
- 5. equipment fault.** Turn 10. Equipment fault has not been established by the setpoint or the comfort report.
- 6. degrees Celsius.** Turn 12. Degrees Celsius supplies the unit attached to the stated target value.
- 7. temperature reading.** Turn 14. Temperature reading is absent from this account; that does not assert that no measurement ever exists.
- 8. service description.** Turn 16. Service description communicates the known complaint without prescribing tests or a control adjustment.
- 9. attributed report.** Turn 18. Attributed report links the comfort account to staff and keeps the target and missing reading separate.
- 10. target temperature.** Turn 20. Target temperature is the intended control value, not proof of actual room conditions or performance.

F Transfer conversation

Page 12

1. setpoint

Manager: "The thermostat setpoint is 22 degrees Celsius."

Setpoint correctly identifies the target rather than a measured ambient temperature.

2. 14:00

Coordinator: "Staff feel warm after 14:00."

The time preserves the supplied afternoon complaint without inventing another period.

3. ambient

Manager: "No confirmed ambient reading is supplied."

Ambient identifies the actual surrounding-air measurement that is missing from the account.

4. unestablished

Coordinator: "The cause remains unestablished."

Neither the setpoint nor the report establishes a technical cause.

02 Briefing and language checks

Identifying equipment and linked components

A Briefing checks / page 13

1. Which label appears in the service request?

C. AC-4

AC-4 is the request label, distinct from the plan and register references.

2. What does the register identify as AH-4?

A. An indoor air handler

The indoor equipment register explicitly identifies an air handler as AH-4.

3. What is unresolved?

D. The relationship between CU-4 and AH-4

The supplied documents do not confirm how the two components are related.

D Language checks / page 18

1. Which summary is accurate?

B. Request AC-4; roof plan CU-4; register AH-4; relationship unverified

The summary preserves the separate source labels and unresolved relationship.

2. What should Owen ask Mira to provide?

D. The relevant existing register information for reconciliation

Existing source records support reconciliation; renaming, pairing by number, or borrowing an unrelated room would add unsupported information.

3. Why is the numeral four insufficient?

B. Similar numbering does not establish verified component relationships

An identifier pattern is not evidence that the recorded components form a confirmed pairing.

4. What should remain visible until clarified?

A. The source of each label and the unverified link

Keeping sources and uncertainty supports accurate reconciliation without losing original information.

02 Conversation answers

Identifying equipment and linked components

E Dialogue gaps

Pages 19-20

- 1. record source.** Turn 2. Record source identifies where each label comes from instead of treating the labels as interchangeable.
- 2. air handler.** Turn 4. Air handler is the indoor item named AH-4 in the register, not a confirmed synonym for CU-4.
- 3. equipment pairing.** Turn 6. Equipment pairing is the relationship that remains unverified despite the shared numeral.
- 4. equipment register.** Turn 8. Equipment register is already available to Mira and can support the document reconciliation.
- 5. service request.** Turn 10. Service request retains the original AC-4 wording while its relationship to other labels is clarified.
- 6. system reference.** Turn 12. System reference may identify the overall arrangement rather than one physical component.
- 7. model number.** Turn 14. Model number identifies a product design or version, unlike the individual-unit serial number.
- 8. served area.** Turn 16. Served area must be established for the identified system rather than borrowed from another request.
- 9. cross-reference.** Turn 18. Cross-reference asks how the documented labels correspond without presuming that any one record is wrong.
- 10. confirmed mapping.** Turn 20. Confirmed mapping supplies verified correspondence before labels are merged or the request is relabeled.

F Transfer conversation

Page 21

1. AC-4

Manager: "The request says AC-4."

AC-4 is the original request label and must remain traceable.

2. CU-4

Coordinator: "The roof plan identifies the outdoor unit as CU-4."

CU-4 belongs to the outdoor item on the roof plan.

3. AH-4

Manager: "The register identifies the air handler as AH-4."

AH-4 is the indoor air-handler reference in the register.

4. unverified

Coordinator: "Their relationship remains unverified."

Similar numbering does not establish a confirmed link between the recorded components.

03 Briefing and language checks

Explaining airflow and distribution reports

03

A Briefing checks / page 22

1. Which opening does the drawing identify as supply?

B. S3

The current drawing explicitly labels S3 as supply air.

2. Where does Tomas report discomfort?

B. Below S3

The client names the area below S3 as the complaint location.

3. What is missing?

C. Airflow measurements

No measured airflow data are supplied with the discomfort report.

D Language checks / page 27

1. Which explanation is accurate?

B. Supply delivers air into the room; return carries air back toward the system

The directional distinction explains the stated functions without changing them into performance findings.

2. What can the drawing labels establish here?

C. The stated functions of S3 and R2

The supplied drawing labels identify functions, not measured operating conditions.

3. Which report invents a measurement?

D. S3 delivers too many liters per second

No airflow measurements support a quantified excess at S3.

4. What is the proper scope of Leila's response?

D. Explain labels and preserve the complaint for assessment

The case supports explanation and accurate reporting, not adjustment or performance conclusions.

03 Conversation answers

Explaining airflow and distribution reports

03

E Dialogue gaps

Pages 28-29

- 1. supply air.** Turn 2. Supply air is the stated delivery function of S3 on the current drawing.
- 2. return air.** Turn 4. Return air describes the flow back toward the system rather than delivery into the room.
- 3. distribution report.** Turn 6. Distribution report records the air-related complaint and its location without adding measurements.
- 4. measured airflow.** Turn 8. Measured airflow would require relevant measurement data, which Tomas has not supplied.
- 5. cubic feet per minute.** Turn 10. Cubic feet per minute is a volume-flow unit, not an air-speed unit.
- 6. Air velocity.** Turn 12. Air velocity is the speed of air movement and must not be confused with volume flow.
- 7. balancing.** Turn 14. Balancing involves technical distribution work and verification, not merely identifying supply and return labels.
- 8. damper.** Turn 16. Damper is an airflow-control component; no adjustment to it is supported by this conversation.
- 9. occupied zone.** Turn 18. Occupied zone is the area used by people, helping locate the comfort issue rather than diagnose equipment.
- 10. performance finding.** Turn 20. Performance finding would require supporting evidence beyond the labels and discomfort account.

F Transfer conversation

Page 30

1. supply

Technician: "S3 is labeled supply air."

Supply is the stated function assigned to S3 by the drawing.

2. return

Client: "R2 is labeled return air."

Return distinguishes R2 from the supply opening named in the complaint.

3. S3

Technician: "The discomfort is reported below S3."

S3 is the specific location identified by the client.

4. measurements

Client: "No airflow measurements have been supplied."

The complaint includes no measured airflow data to support a performance conclusion.

04 Briefing and language checks

Checking equipment ratings and refrigerant references

04

A Briefing checks / page 31

1. What does proposal A's 5 kilowatts describe?

C. Cooling capacity

Proposal A explicitly labels the value as cooling capacity.

2. What does proposal B's 5 kilowatts describe?

D. Electrical input

Proposal B labels the value electrical input, which is a different quantity.

3. What conclusion is supported?

C. The numbers alone cannot establish equivalent systems

The matching unit and value do not make the two differently labeled quantities equivalent.

D Language checks / page 36

1. Which comparison is valid from the supplied entries?

C. A's value is cooling capacity; B's is electrical input

Only the differently labeled quantities can be identified; performance equivalence is not established.

2. What additional information is needed?

C. Full data sheets, rating conditions, and component pairings

Matching quantities, conditions, and actual equipment combinations are needed; headline numbers or refrigerant names do not fill those gaps.

3. Which conclusion exceeds the evidence?

A. The systems have equal efficiency

The two entries do not provide the corresponding output and input data needed for that claim.

4. What does a refrigerant designation provide?

C. Identification to be checked against the actual equipment specification

The designation identifies the fluid but does not establish compatibility or authorization.

04 Conversation answers

Checking equipment ratings and refrigerant references

E Dialogue gaps

Pages 37-38

- 1. cooling capacity.** Turn 2. Cooling capacity identifies A's heat-removal output rating, not its electrical power input.
- 2. Electrical input.** Turn 4. Electrical input is the power required by the equipment, not the useful cooling output.
- 3. efficiency.** Turn 6. Efficiency comparison needs corresponding output and input information on a consistent basis.
- 4. data sheet.** Turn 8. Data sheet supplies product characteristics beyond the two incomplete proposal entries.
- 5. equipment combination.** Turn 10. Equipment combination identifies the components whose joint specifications are relevant to the proposal.
- 6. refrigerant designation.** Turn 12. Refrigerant designation identifies the specified fluid but does not establish system equivalence or replacement approval.
- 7. substitution review.** Turn 14. Substitution review evaluates a proposed replacement and is not bypassed by a familiar refrigerant name.
- 8. Refrigerant charge.** Turn 16. Refrigerant charge concerns the quantity of working fluid, distinct from capacity and refrigerant identity.
- 9. like-for-like comparison.** Turn 18. Like-for-like comparison uses matching quantities and conditions instead of the superficially matching five-kilowatt figures.
- 10. selection basis.** Turn 20. Selection basis is the supporting information still needed before an equipment choice can be approved.

F Transfer conversation

Page 39

1. capacity

Client: "A gives 5 kilowatts of cooling capacity."

Capacity identifies the cooling-output quantity stated in proposal A.

2. input

Estimator: "B gives 5 kilowatts of electrical input."

Input identifies power required by the equipment, not cooling output.

3. sheets

Client: "We need complete data sheets and pairings."

Full data sheets and component pairings are missing from the supplied proposals.

4. approved

Estimator: "No selection is yet approved."

The comparison does not establish an accepted equipment choice or authorization.

05 Briefing and language checks

Reporting moisture, symptoms, and service findings

A Briefing checks / page 40

1. What did Elena report yesterday?

D. Water near AH-6 at 15:00

Elena's account concerns observed water near AH-6 at the stated time.

2. What does today's visit note establish?

D. The nearby floor was dry and the source was not identified

The note records a later surface condition and an unresolved source.

3. What does the earlier drain-pan entry prove?

A. Only that a concern was recorded, not a causal link

The historical concern is not linked to the water by the supplied evidence.

D Language checks / page 45

1. Which summary preserves chronology?

D. Water near AH-6 yesterday at 15:00; floor dry today; source unidentified

The accurate summary retains near AH-6, yesterday's 15:00 report, today's dry floor, and the unidentified source without changing attribution or inventing cause.

2. Which wording improperly asserts origin?

C. Water came from AH-6

The supplied report gives proximity, not proof that AH-6 produced the water.

3. How should the drain-pan entry be handled?

C. Retain it as history with no established causal link

Relevant history can be preserved without asserting a cause or resolution.

4. What can the dry-floor note establish about repair?

B. No completed repair by itself

A surface observation is not evidence that corrective work occurred.

05 Conversation answers

Reporting moisture, symptoms, and service findings

E Dialogue gaps

Pages 46-47

1. chronology. Turn 2. Chronology keeps yesterday's report and today's observation separate instead of treating them as simultaneous claims.

2. visit note. Turn 4. Visit note is the source for today's dry-floor observation, not yesterday's client report.

3. drain-pan concern. Turn 6. Drain-pan concern is historical information, not a confirmed cause of the later water report.

4. evidence boundary. Turn 8. Evidence boundary limits the account to observed location rather than asserting origin from AH-6.

5. historical entry. Turn 10. Historical entry must retain its recorded status as a concern rather than being upgraded to confirmed failure.

6. source identification. Turn 12. Source identification remains unresolved because today's note explicitly says the source was not identified.

7. repair status. Turn 14. Repair status cannot be inferred from a later dry surface when no corrective work is documented.

8. recurrence history. Turn 16. Recurrence history requires evidence of repeated events, not merely an earlier concern about a component.

9. causal link. Turn 18. Causal link is the missing connection between the earlier concern and the reported water.

10. record reconciliation. Turn 20. Record reconciliation combines the three sources while preserving time, attribution, and uncertainty.

F Transfer conversation

Page 48

1. 15:00

Client: "Water was reported near AH-6 at 15:00 yesterday."

The time belongs to Elena's report on the previous day.

2. dry

Coordinator: "Today the nearby floor was recorded as dry."

Dry describes the surface observed during the later visit.

3. concern

Client: "The earlier log mentions a drain-pan concern."

Concern preserves the historical wording without upgrading it to a confirmed failure.

4. identified

Coordinator: "The source was not identified."

The visit note leaves the origin of the reported water unresolved.

05

06 Briefing and language checks

Discussing refrigeration alarms and temperature records

A Briefing checks / page 49

1. What was recorded at 06:10?

C. A cabinet display value of 8 degrees Celsius

The record supplies a cabinet display value, not product temperatures or duration.

2. What do the matching timestamps establish?

B. The two entries share a recorded time

Coincident timestamps do not establish cause, duration, or product safety.

3. Who must decide the food-safety question?

A. The responsible food-safety lead under site procedures

The scenario assigns that decision to the responsible lead using the site's procedures.

D Language checks / page 54

1. Which summary belongs in the escalation?

C. Display 8 degrees Celsius at 06:10; product data absent; cause and duration unconfirmed

The summary preserves the actual value and all material information limits.

2. Why is defrost caused it unsupported?

D. The shared timestamp does not establish causation

Coincident entries show a shared recorded time, not a verified cause, completed cycle, duration, or food-temperature history.

3. Who should receive the food-release question?

D. The responsible food-safety lead using site procedures

Product decisions require the assigned responsible process rather than an improvised technical assurance.

4. Which statement improperly fills a data gap?

C. The food was warm for only one minute

No event duration or product exposure period is established by the record.

06 Conversation answers

Discussing refrigeration alarms and temperature records

E Dialogue gaps

Pages 55-56

- 1. display value.** Turn 2. Display value is equipment information and does not supply the missing product temperatures or exposure history.
- 2. product temperature.** Turn 4. Product temperature concerns the food itself and is absent from the information supplied.
- 3. timestamp.** Turn 6. Timestamp establishes the recorded time shared by the entries, not why the alarm occurred.
- 4. event duration.** Turn 8. Event duration remains unknown because one timestamp does not establish a complete interval.
- 5. food-safety lead.** Turn 10. Food-safety lead is the assigned decision-maker for product safety under the site's procedures.
- 6. scheduled defrost.** Turn 12. Scheduled defrost is a recorded event sharing the time, not a proven explanation for the alarm.
- 7. Alarm clearance.** Turn 14. Alarm clearance concerns equipment alarm status and does not establish product safety or past exposure.
- 8. exposure history.** Turn 16. Exposure history describes conditions experienced by the products over time, which these isolated records do not establish.
- 9. escalation summary.** Turn 18. Escalation summary passes the actual records and missing information to the responsible lead without authorizing release.
- 10. Product disposition.** Turn 20. Product disposition is the responsible decision about the food, distinct from interpreting the equipment records.

F Transfer conversation

Page 57

1. 8

Manager: "The display recorded 8 degrees Celsius."

Eight is the cabinet display value, not a verified product temperature.

2. 06:10

Technician: "The timestamp was 06:10."

The timestamp records one time and does not establish event duration.

3. supplied

Manager: "Product temperatures were not supplied."

The absence of product readings is a material limit on the information.

4. lead

Technician: "The responsible food-safety lead must decide under site procedures."

The assigned lead makes the product decision rather than the equipment display alone.

07 Briefing and language checks

Coordinating ductwork, access, and changes

A Briefing checks / page 58

1. Why does Priya want the grille moved?

D. It visually conflicts with a pendant light

The request concerns appearance, not an established system fault.

2. Which consequences have been assessed?

A. None of the listed change effects

Ductwork, restoration, access, price, and timing remain unassessed.

3. What are the stated choices?

C. Retain the reviewed layout or request a coordinated change proposal

The brief supplies two bounded options without authorizing an immediate alteration.

D Language checks / page 63

1. Which response explains the choice accurately?

C. Retain the reviewed layout or request a coordinated proposal with impacts assessed

The response offers the two supplied options and preserves the need to assess consequences.

2. Which item must not disappear from review?

B. Service access

Service access is explicitly one of the unassessed matters in the brief.

3. What does requesting a proposal authorize here?

A. Assessment of the change, not its installation

A request for review does not equal agreement to carry out the proposed work.

4. Which phrase overstates certainty?

D. We can move it with no cost or timing effect

Cost and timing effects remain unknown, so a no-impact promise is unsupported.

07 Conversation answers

Coordinating ductwork, access, and changes

E Dialogue gaps

Pages 64-65

1. design preference. Turn 2. Design preference identifies the appearance concern without claiming a performance fault.

2. visual conflict. Turn 4. Visual conflict describes the relationship in appearance between the grille and pendant light.

3. ductwork. Turn 6. Ductwork is one of the hidden systems whose changes have not been assessed for the proposed move.

4. Ceiling restoration. Turn 8. Ceiling restoration concerns reinstating affected finishes and remains part of the unassessed change scope.

5. access provision. Turn 10. Access provision concerns how the relevant services remain reachable after a proposed layout change.

6. reviewed layout. Turn 12. Reviewed layout is the existing arrangement available as the first stated option.

7. change proposal. Turn 14. Change proposal is the requested assessed alternative, not authorization to install or acceptance of its terms.

8. schedule impact. Turn 16. Schedule impact remains unassessed, so unchanged completion cannot be guaranteed.

9. cost impact. Turn 18. Cost impact must be assessed with the defined work rather than assumed absent because the visible move seems small.

10. decision record. Turn 20. Decision record preserves the selected assessment step without turning it into an instruction to proceed.

F Transfer conversation

Page 66

1. layout

Coordinator: "You can retain the reviewed layout."

Layout refers to the existing reviewed arrangement available as the first option.

2. proposal

Client: "Or request a coordinated change proposal."

Proposal is the assessed alternative the client may request before deciding.

3. assessment

Coordinator: "Cost and timing still need assessment."

The effects on price and schedule have not yet been evaluated.

4. relocation

Client: "This request does not authorize relocation."

Requesting a proposal does not approve moving the supply grille.

08 Briefing and language checks

Handing over commissioning and maintenance information

A Briefing checks / page 67

1. What does the commissioning report list?

D. A deferred seasonal check

The report explicitly identifies a check that has been deferred rather than completed.

2. What responsibility does Nia accept?

B. Document correction and a Friday update

Nia's commitment is to document correction and communication, not the unassigned scheduling task.

3. What remains unresolved about the seasonal check?

C. Both the date and the scheduling owner

The check is unbooked and responsibility for arranging it is not confirmed.

D Language checks / page 72

1. Which summary preserves all responsibilities?

B. Nia corrects documents and updates Friday; seasonal-check owner and date remain open

The summary keeps Nia's accepted task distinct from the unresolved scheduling responsibility.

2. What does Friday identify?

B. Nia's promised update

The only Friday commitment is communication about follow-up.

3. Which inference is unsupported?

B. Correcting the maintenance sheet completes the seasonal check

A document correction is not performance of the deferred verification.

4. What should the handover keep open?

C. Correct model guidance, deferred verification, scheduling owner, and date

The remaining items are separate; receipt, a document task, and an update commitment do not complete verification, assign scheduling ownership, or book a check.

08 Conversation answers

Handing over commissioning and maintenance information

E Dialogue gaps

Pages 73-74

- 1. document mismatch.** Turn 2. Document mismatch identifies the wrong-model maintenance sheet as a correction task.
- 2. deferred check.** Turn 4. Deferred check remains outstanding even though its existence is recorded in the delivered report.
- 3. scheduling owner.** Turn 6. Scheduling owner has not been confirmed, so the arranging responsibility remains open.
- 4. ownership gap.** Turn 8. Ownership gap names the unassigned scheduling responsibility rather than silently giving it to Nia.
- 5. update commitment.** Turn 10. Update commitment is Nia's promise to communicate on Friday, not a promised technical-check result.
- 6. model-specific guidance.** Turn 12. Model-specific guidance links the corrected maintenance information to the actual equipment rather than a similar model.
- 7. performance result.** Turn 14. Performance result cannot be generalized to all conditions when a seasonal check is still deferred.
- 8. warranty terms.** Turn 16. Warranty terms retain their actual meaning and are not revised by the document update or deferred-check discussion.
- 9. responsibility readback.** Turn 18. Responsibility readback distinguishes the assigned document task from the unassigned seasonal scheduling task.
- 10. handover register.** Turn 20. Handover register should record receipt alongside corrections and outstanding verification without claiming full completion.

F Transfer conversation

Page 75

1. Nia

Client: "Nia owns document correction."

Nia accepts the correction task, not confirmed ownership of arranging the seasonal check.

2. Friday

Coordinator: "The update is promised for Friday."

Friday is the communication commitment rather than a booked technical appointment.

3. unbooked

Client: "The seasonal check is still unbooked."

No confirmed date has been supplied for the deferred seasonal verification.

4. confirmed

Coordinator: "Its scheduling owner is not yet confirmed."

The responsibility for arranging the check remains an explicit open question.

09 Conversation answers

Total cooling hides the split

E Extended conversation

Pages 76-77

1. sensible load. Turn 2. The supplied ten-kilowatt sensible load concerns temperature change; it must be compared with sensible capacity, not total capacity alone.

2. latent load. Turn 4. The two-kilowatt latent load concerns the moisture-removal part of this cooling duty; it is not a temperature value in degrees.

3. total capacity. Turn 6. Twelve kilowatts total equals nine sensible plus three latent here; extra latent capacity does not erase the stated sensible-capacity shortfall.

4. sensible heat ratio. Turn 8. SHR is sensible capacity divided by total capacity, nine divided by twelve or 0.75 for A; it is not output divided by electrical input.

5. comparison basis. Turn 10. The case gives consistent net capacities at the same conditions; B's 10.4 sensible and 2.6 latent exceed the supplied ten and two loads respectively.

6. load profile. Turn 12. Different combinations of temperature and moisture loads require different comparisons; the highest sensible fraction is not automatically best for every space.

F Transfer exchange

Page 78

1. 1.6

Manager: C's latent capacity is 1.6 kW.

Eight minus 6.4 equals 1.6 kW latent, below the supplied two-kilowatt latent load.

2. 0.4

Estimator: C therefore has a latent shortfall of 0.4 kW.

Two minus 1.6 equals a 0.4 kW shortfall despite C matching the eight-kilowatt total load.

3. D

Manager: Option D covers both supplied component loads numerically.

D supplies 6.3 sensible and 2.7 latent, above six and two respectively; this is not installation approval.

4. 0.7

Estimator: D's sensible heat ratio is 0.7 .

6.3 divided by nine equals 0.7; SHR is a capacity split, not electrical efficiency.

Background reading: [Trane TRACE: total cooling capacity, sensible heat ratio, and rating basis](#). The conversation is an original fictional teaching case.

10 Conversation answers

Circulation is not all outdoor air

E Extended conversation

Pages 79-80

1. total supply. Turn 2. Six hundred cubic meters per hour divided by a three-hundred-cubic-meter room gives two total-supply air changes per hour, including recirculated air.

2. outdoor-air flow. Turn 4. Only 150 cubic meters per hour is outdoor air in this case; the other 450 is recirculated, not an additional outdoor-air contribution.

3. air changes per hour. Turn 6. ACH divides a consistently expressed hourly airflow by room volume; the total-supply result is two and the outdoor-air result is 0.5.

4. outdoor-air fraction. Turn 8. Outdoor flow divided by total supply is $150/600 = 0.25$, or 25 percent; this denominator is flow, not room volume.

5. time basis. Turn 10. A cubic meter contains one thousand liters and an hour has 3,600 seconds; 600 cubic meters per hour is about 166.7 liters per second.

6. air mixing. Turn 12. An air-change quantity describes delivered volume relative to room volume; it does not guarantee complete replacement of every portion of room air in a fixed interval.

F Transfer exchange

Page 81

1. three

Coordinator: Total-supply air changes per hour are three .

720 divided by 240 equals three total-supply air changes per hour, including the outdoor portion.

2. 0.75

Specialist: Outdoor-air changes per hour are 0.75 .

180 divided by 240 equals 0.75 outdoor-air changes per hour, not three.

3. 25

Coordinator: Outdoor air forms 25 percent of the supply.

180 divided by 720 times one hundred equals twenty-five percent.

4. 50

Specialist: Outdoor flow is 50 liters per second.

180 cubic meters per hour times one thousand divided by 3,600 equals fifty liters per second.

10

Background reading: [CDC/NIOSH: airflow, air changes, filtration, and mixing limitations](#). The conversation is an original fictional teaching case.

11 Conversation answers

Use the right saturation reference

E Extended conversation

Pages 82-83

1. dew point. Turn 2. The supplied vapor-side dew point is 5 C; superheat is referenced to saturated vapor at the matching pressure, not the zero-degree bubble point.

2. superheat. Turn 4. Twelve degrees Celsius vapor temperature minus the five-degree dew point gives a seven-kelvin temperature difference; the original temperature reading remains twelve Celsius.

3. bubble point. Turn 6. The supplied liquid-side bubble point is 40 C; it is the saturated-liquid reference for subcooling at that reading's pressure.

4. subcooling. Turn 8. Forty degrees Celsius bubble point minus thirty-five degrees Celsius liquid temperature gives five kelvin subcooling; using dew would incorrectly give ten.

5. temperature glide. Turn 10. A blend with temperature glide has distinct bubble and dew temperatures at a given pressure; the supplied reference pairs differ by five kelvin.

6. matching pressure. Turn 12. The vapor and liquid readings have different pressure-specific reference pairs; a reference from the other location cannot be substituted into the calculation.

F Transfer exchange

Page 84

1. dew

Coordinator: The vapor calculation uses the dew point.

Superheat uses the saturated-vapor dew reference at the corresponding pressure, not the bubble value.

2. seven

Technician: The superheat is seven kelvin.

Eleven minus four equals seven kelvin; using minus one would incorrectly give twelve.

3. bubble

Coordinator: The liquid calculation uses the bubble point.

Subcooling uses the saturated-liquid bubble reference at the matching pressure, not the dew value.

4. six

Technician: The subcooling is six kelvin.

Thirty-seven minus thirty-one equals six kelvin; forty-two minus thirty-one would use the wrong reference.

11

Background reading: [Parker Sporlan: dew and bubble references for refrigerant blends](#). The conversation is an original fictional teaching case.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Clarifying comfort complaints and thermostat readings

Ask what the number means: Is 22 the setpoint or a measured room temperature?

Acknowledge the people: Staff report feeling warm in the meeting room.

Avoid assigning a failed part: No sensor fault has been established.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Identifying equipment and linked components

Quote the request: The service request calls the system AC-4.

Preserve source attribution: Each label should stay attached to its source.

Request reconciliation: We need a confirmed cross-reference between these entries.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Explaining airflow and distribution reports

Name the two functions: S3 is supply air; R2 is return air.

Attribute the labels: Those are the functions shown on the current drawing.

Name common units: Volume flow may be stated in cubic feet per minute or liters per second.

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Checking equipment ratings and refrigerant references

Ask the key question: Five kilowatts of which quantity?

Name the distinction: Cooling output and electrical input are different.

Keep conditions attached: Compare ratings under the relevant stated conditions.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Reporting moisture, symptoms, and service findings

State yesterday's account: The client reported water near AH-6 at 15:00 yesterday.

Name the unresolved issue: The source was not identified.

Avoid changing concern to failure: The record says concern, not confirmed failure.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Discussing refrigeration alarms and temperature records

Quote the actual record: The cabinet display recorded 8 degrees Celsius at 06:10.

Identify the coincident entries: The alarm and scheduled-defrost entries share a timestamp.

Name the decision route: The responsible food-safety lead must decide under the site procedures.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Coordinating ductwork, access, and changes

Acknowledge the preference: I understand why you want the grille clear of the pendant light.

Identify the unassessed items: Ductwork, restoration, access, price, and timing still need assessment.

Keep the choice bounded: Those are the two decisions available at this stage.

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Handing over commissioning and maintenance information

Acknowledge the report: You have received the commissioning report.

Accept a specific task: I will take responsibility for document correction.

State the ownership gap: Responsibility for arranging that check is not confirmed.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

acceptance criterion	identify the acceptance criterion	69
access provision	review the access provision	59
afternoon pattern	record the afternoon pattern	5
air changes per hour	distinguish total and outdoor-air changes	24
air distribution	describe air distribution	23
air handler	identify the air handler	14
air velocity	distinguish air velocity	24
airflow rate	report the airflow rate	23
alarm acknowledgment	distinguish alarm acknowledgment	51
alarm clearance	distinguish alarm clearance	51
alarm history	review the alarm history	50
ambient temperature	record ambient temperature	5
asset hierarchy	clarify the asset hierarchy	15
asset tag	preserve the asset tag	14
attributed report	retain an attributed report	6
balancing	distinguish balancing from explanation	24
blend	identify a refrigerant blend	33
branch duct	review the branch duct	59
cabinet display	quote the cabinet display	50
causal link	establish a causal link	41
causation	avoid assuming causation	50
ceiling restoration	assess ceiling restoration	59
ceiling void	coordinate the ceiling void	59
change proposal	request a change proposal	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

chronology preserve the chronology	41
closeout status state the closeout status	69
coefficient of performance interpret the coefficient of performance	32
coincident entries compare coincident entries	50
comfort complaint log a comfort complaint	5
commissioning distinguish commissioning from document delivery	68
commissioning report receive the commissioning report	68
compatibility verify compatibility	33
component reference retain the component reference	14
condensate identify a condensate reference	41
condensate line identify the condensate line	41
condensation distinguish condensation from a diagnosis	42
condensing unit verify the condensing-unit reference	14
confirmed mapping obtain confirmed mapping	15
control schedule review the control schedule	6
cooling capacity compare cooling capacity	32
coordination review arrange a coordination review	60
correction owner name the correction owner	68
cost impact assess the cost impact	60
cross-reference establish a cross-reference	15
cubic feet per minute state cubic feet per minute	23
damper identify the damper	24
data gap identify the data gap	51
data sheet obtain the complete data sheet	32

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

decision record confirm the decision record	60
deferral reason preserve the deferral reason	69
deferred check record the deferred check	68
defrost identify the defrost entry	50
degrees Celsius state the value in degrees Celsius	5
design airflow compare design airflow	24
design preference record the design preference	60
diffuser identify the diffuser	23
display value record the display value	50
displayed value clarify the displayed value	6
distribution report clarify the distribution report	24
document mismatch identify the document mismatch	68
document receipt acknowledge document receipt	69
draft describe a reported draft	24
drain pan record a drain-pan concern	41
drainage path identify the drainage path	42
dry observation qualify the dry observation	42
duct identify the duct	23
duct connection assess the duct connection	59
ductwork review the ductwork	23
efficiency compare efficiency on a consistent basis	32
electrical input identify electrical input	32
equipment combination verify the equipment combination	32
equipment fault avoid assuming an equipment fault	6

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

equipment pairing	verify the equipment pairing	14
equipment register	consult the equipment register	14
escalation summary	give an escalation summary	51
event duration	establish event duration	50
evidence boundary	preserve the evidence boundary	42
exposure history	establish exposure history	51
finding	state the actual finding	42
flexible duct	identify the flexible duct	59
food-safety lead	notify the food-safety lead	51
handover register	update the handover register	69
heat gain	assess heat gain	6
historical entry	distinguish a historical entry	41
HVAC	discuss the HVAC system	5
identification query	raise an identification query	15
indoor unit	identify the indoor unit	14
instruction to proceed	distinguish an instruction to proceed	60
intermittent symptom	describe an intermittent symptom	42
kilowatt	state the rating in kilowatts	32
kilowatt-hour	distinguish a kilowatt-hour	32
label discrepancy	record the label discrepancy	15
latent cooling	account for latent cooling	6
leakage	avoid assuming leakage source	42
like-for-like comparison	make a like-for-like comparison	33
liters per second	record liters per second	23

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

maintenance requirement	verify the maintenance requirement	69
maintenance sheet	correct the maintenance sheet	68
manufacturer specification	consult the manufacturer specification	33
matched system	identify the matched system	32
measured airflow	record measured airflow	24
measured value	distinguish the measured value	5
model number	confirm the model number	15
model-specific guidance	obtain model-specific guidance	68
moisture report	retain the moisture report	41
nameplate	use recorded nameplate information	15
nearby floor	describe the nearby floor	41
nominal capacity	distinguish nominal capacity	32
occupancy	describe room occupancy	6
occupant feedback	preserve occupant feedback	5
occupied zone	identify the occupied zone	24
operating condition	state the operating conditions	32
operating record	retain the operating record	69
outdoor unit	identify the outdoor unit	14
outstanding verification	track outstanding verification	69
overflow	distinguish a reported overflow	42
ownership gap	preserve the ownership gap	68
pendant light	coordinate the pendant light	59
performance finding	distinguish a performance finding	24
performance result	distinguish a performance result	69

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

product disposition determine product disposition	51
product temperature distinguish product temperature	50
rating condition check the rating conditions	32
record custodian identify the record custodian	15
record reconciliation complete record reconciliation	42
record source identify the record source	14
records reconciliation complete records reconciliation	15
recurrence history verify recurrence history	42
reflected ceiling plan review the reflected ceiling plan	59
refrigerant identify the specified refrigerant	33
refrigerant charge distinguish refrigerant charge	33
refrigerant designation preserve the refrigerant designation	33
refrigerated cabinet identify the refrigerated cabinet	50
register distinguish the register	23
relative humidity report relative humidity	6
release decision refer the release decision	51
repair status verify repair status	42
reported period preserve the reported period	6
reported water locate the reported water	41
responsibility readback give a responsibility readback	69
return air identify return air	23
return grille identify the return grille	23
reviewed layout retain the reviewed layout	60
revised layout review the revised layout	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

rigid duct distinguish rigid duct	59
roof plan cite the roof plan	14
schedule impact assess the schedule impact	60
scheduled defrost distinguish scheduled defrost	50
scheduling owner confirm the scheduling owner	68
scope impact identify the scope impact	60
seasonal check schedule the seasonal check	68
selection basis establish the selection basis	33
sensible cooling distinguish sensible cooling	5
sensible heat ratio calculate the sensible heat ratio	33
sensor identify the relevant sensor	5
sensor location identify the sensor location	51
serial number record the serial number	15
served area confirm the served area	14
service clearance preserve service clearance	59
service description clarify the service description	6
service log consult the service log	41
service request reconcile the service request	15
setpoint distinguish the setpoint	5
site procedure follow the site procedure	51
source identification keep source identification open	41
spread distinguish spread from throw	24
subcooling distinguish subcooling from superheat	33
substitution review request a substitution review	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

superheat report superheat in kelvin	33
supply air identify supply air	23
supply grille locate the supply grille	23
supply grille relocation assess a supply grille relocation	59
system reference clarify the system reference	14
target temperature state the target temperature	6
technical cause keep the technical cause open	51
technical feasibility establish technical feasibility	60
temperature excursion assess a recorded temperature excursion	50
temperature reading identify the temperature reading	5
test condition record the test condition	69
thermostat identify the thermostat	5
throw review diffuser throw	24
timestamp preserve the timestamp	50
trend record review the trend record	51
unassessed consequence keep unassessed consequences visible	60
unbooked check retain the unbooked check	68
unresolved concern retain an unresolved concern	42
unverified link retain an unverified link	15
update commitment honor the update commitment	68
visit note quote the visit note	41
visual conflict describe the visual conflict	59
warranty terms preserve warranty terms	69
zone identify the relevant zone	6

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] US Department of Energy. Weatherization Training Glossary.

[Open the original source](#)

Background terminology for air distribution, temperature, humidity, and building systems. Historical program rules and technical procedures are not taught here. Checked 10 October 2026.

[2] Carrier. Return Air Vent System.

[Open the original source](#)

Reference for distinguishing supply and return airflow. No airflow targets, maintenance intervals, adjustments, or product recommendations are adopted. Checked 10 October 2026.

[3] US Environmental Protection Agency. Section 608 Technician Certification.

[Open the original source](#)

Context for professional boundaries around refrigerant work. The book does not teach certification requirements or authorize servicing. Checked 10 October 2026.

[4] US Food and Drug Administration. Refrigerator Thermometers: Cold Facts about Food Safety.

[Open the original source](#)

Background for treating temperature information carefully. The fictional cabinet case supplies no food-release decision or substitute for site procedures. Checked 10 October 2026.

Phil Stilwell

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